

## 完成数据科学作业 (43 条)

<https://chatgpt.com/c/6a949e98-9268-83e8-8c57-063412392c64>

21

用户消息

22

AI 消息

24

图片

0

代码块

0

表格

0

链接

### 关键词

churn data model customer customers text logistic print part recall  
dataset regression python threshold tenure readme totalcharges science  
feature notebook analysis accuracy precision telco github assignment  
test interpretation

### 大纲 · 21

- #1 以下是我将要完成的任务 请协助我完成它Assignment 1 - Do data science with a chat or coding assistance end2end Due Tuesday by 11:59pm Poi...
- #4 没问题，我今天先完成part1，后面还有大约48小时可以完成part2，这是老师给的文档内容
- #6 是在这里下载这个文件么
- #8 下载好了，是这份文件么
- #10 这些内容我是否需要编辑进入一个doc文本
- #12 他需要订阅50美金/月，这是必须的么
- #14 我已准备好，我们开始吧
- #16 [Image]
- #18 [Image]
- #20 [Image]
- #22 [Image]
- #24 [Image]
- #26 是的，我暂离了一段时间，现在我放到正确的文件夹内了，我们可以继续了
- #28 [Image] [Image] [Image]
- #30 是的，你推测的没错，这是之前运算的结果，以及后续的
- #32 [Image] [Image] [Image]
- #34 [Image] [Image]
- #36 [Image] [Image] [Image] [Image]
- #38 全部运行成功没有error，以及这是文件
- #40 好的，我们开始github部分的任务吧

#42 已创建/上传，现在最外侧有一个part1文件夹和一个readme，part1文件夹中有上传的两个文件和一个readme

#1 U 你

以下是我将要完成的任务 请协助我完成它Assignment 1 - Do data science with a chat or coding assistance end2end

Due Tuesday by 11:59pm Points 500 Submitting a website url Available Aug 26 at 12am - Sep 1 at 11:59pm

Assignment details:

Part 1:

<https://docs.google.com/document/d/1NMI64Cxlyq3BznMgAFotEsp0fHpEkQe4b7Oi98SEjaA/edit?tab=t.0>Links to an external site.

You can use any coding assistant or chatbot which supports agentic coding (like chatgpt or codex)

Two deliverables:

Make sure you have all artifacts captured in a nice README.md file in your github directory - make sure you organize it as assignment 1

Also upload any other artifacts in the directory of github

Make sure you give public view permission for the github

Also do a youtube video highlighting the end2end journey and read and understand and provide a paraphrase of the output

make sure you upload youtube video and provide the link to it in README.md

Part 2:

Replicate the data science experiments i did in your favorite coding assistant - i provided the prompts in the repository

<https://github.com/dlmastery/datascienceexamples>Links to an external site.

make sure you publish them in github repo and walk thru these in a youtube video and upload and link to README.md

Note: use your creativity and not need to do exactly like above - improvise if possible. the minimum bar is above

thanks

vijay eranti

#2 C CHATGPT

可以，我会按“最终能直接交作业”的目标协助你，而不只是解释要求。第一步我会先读老师给的 Google Doc 具体 Part 1 内容，再检查 GitHub 示例仓库里的 prompts，然后给你整理成可执行的目录结构、实验步骤、README 内容和 YouTube 演示脚本。

可以，我可以陪你把这个 Assignment 1 从头做到能提交，包括 **代码、实验结果、GitHub 目录、README、结果解释/paraphrase，以及 YouTube 演示稿。**

我已经先确认了你贴出来的要求。这个作业本质上分成两个部分：

#### 1. Part 1: **自己完成一个 end-to-end Data Science 项目**

- 可以使用 ChatGPT、Codex 或其他 agentic coding assistant。
- 不只是写代码，而是要展示完整的数据科学流程。
- 所有 artifacts 放进 GitHub。
- GitHub repo 必须公开。
- README 要整理得清楚，并放 YouTube 视频链接。
- 视频中需要展示完整 journey，并且你要**理解结果，然后用自己的话解释结果。**

#### 1. Part 2: **复现老师 GitHub repository 里的 Data Science experiments**

- 老师给的是 `dlmastery/data_science_examples`。
- 按 repository 中提供的 prompts，在你选择的 coding assistant 里重新做实验。
- 把代码、结果、prompt、截图/图表等放 GitHub。
- 再录 YouTube walkthrough。
- YouTube 链接继续放 README。

老师还特别说了：**不需要 100% 原样复制，可以发挥 creativity，只要最低要求全部满足。**

目前我能确认 Google Doc 的标题是 “**cmpe 255 assignment 1 fall 2025**”，但 Google Docs 页面本身没有把正文开放给当前网页读取，所以我暂时还看不到 Part 1 文档里面的数据集/具体实验说明。

□cite□turn411051view0□

你给的 GitHub 用户 `dlmastery` 是存在的，而且目前公开有多个 repositories。

□cite□turn542232search2□ 但指定的 `data_science_examples` repo 当前网页索引没有成功返回内容，所以后面如果你把 repository 文件下载下来上传给我，或者把里面 prompts 发给我，我可以逐个直接做。

**我建议我们最终把你的 GitHub 做成下面这种结构**

```
CMPE255/
├── Assignment_1/
│   ├── README.md
│   ├── Part_1_End_to_End_Data_Science/
│   │   ├── README.md
│   │   ├── data/
│   │   ├── notebooks/
│   │   │   └── end_to_end_data_science.ipynb
│   │   ├── src/
│   │   ├── images/
│   │   │   ├── distribution.png
│   │   │   ├── correlation.png
│   │   │   ├── confusion_matrix.png
│   │   │   └── feature_importance.png
│   │   └── results/
│   └── Part_2_Data_Science_Experiments/
│       ├── README.md
│       ├── experiment_01/
│       ├── experiment_02/
│       ├── experiment_03/
│       └── ...
```

这样比把所有 notebook 丢在一个目录里专业很多，也符合老师所说的 “make sure you organize it as assignment 1”。

## Part 1 我会带你完成这样的完整 Data Science Pipeline

```
Problem Definition
↓
Dataset Selection
↓
Load Data
↓
Data Understanding
↓
Data Cleaning
↓
Exploratory Data Analysis
↓
Visualization
↓
Feature Engineering
↓
Train/Test Split
↓
Baseline Model
↓
Model Training
↓
Model Evaluation
↓
Model Comparison
↓
Feature Importance / Interpretation
↓
Final Conclusion
```

这里最重要的一点是：**不要只让 AI 生成代码然后结束。**

老师要求你 “read and understand and provide a paraphrase of the output”，所以我们的 notebook 会刻意做成：

```
AI Prompt
↓
Generated / Assisted Code
↓
Output
↓
My Interpretation
```

例如：

```
Accuracy = 0.91
```

你的视频里不能只说：

```
"The accuracy is 91%."
```

而应该解释成类似：

*"The model correctly classified approximately 91% of the samples in the test dataset. However, accuracy alone may not be sufficient, especially if the dataset contains class imbalance, so I also examined precision, recall, F1-score, and the confusion matrix."*

这样老师一看就知道你**真的理解模型输出**。

## **Part 1 的选题我建议不要搞得太复杂**

如果老师 Google Doc 没有限定 dataset, 我比较推荐你做:

### **Customer Churn Prediction**

原因是它能覆盖很多 data science 内容:

```
Classification
EDA
Missing values
Categorical variables
Feature engineering
Correlation
Logistic Regression
Decision Tree / Random Forest
Confusion Matrix
Precision / Recall / F1
Feature Importance
```

这样视频也很好讲。

另一种非常适合作业的是:

### **Titanic Survival Prediction**

优点是简单、安全、不容易踩环境问题, 但是缺点是比较普通。

如果你希望这个 Assignment 看起来比最低要求更好, 我会更倾向:

### **Customer Churn + multiple model comparison**

例如:

```
Logistic Regression
Random Forest
Gradient Boosting
```

然后比较:

```
Accuracy
Precision
Recall
F1
ROC-AUC
```

最后回答:

*Which model performed best and why?*

这个非常像一个完整的数据科学 mini project。

---

## **Part 2 不要机械复制，我建议做成“Prompt → AI → Experiment → Interpretation”**

老师给的实验核心其实是在展示 AI-assisted data science。

公开类似的数据科学 prompt repositories 中通常包含例如：

- classification
- AutoML
- hyperparameter tuning
- EDA
- synthetic data generation
- imbalance handling
- feature importance
- visualization
- LIME / SHAP
- time-series analysis

这一类数据科学任务。□cite□turn542232search0□turn542232search1□

我们可以把每个 experiment 写成：

```
## Experiment 1 – Exploratory Data Analysis
```

```
### Original Prompt
```

```
...
```

```
### Tool Used
```

```
ChatGPT / Codex
```

```
### Dataset
```

```
...
```

```
### Generated Approach
```

```
...
```

```
### Code
```

```
...
```

```
### Output
```

```
...
```

```
### My Interpretation
```

```
...
```

```
### What I Learned
```

```
...
```

这一点会让你的 repo 看上去比“直接上传 notebook”完整很多。

---

## 最终 README 我会帮你做到这种级别

后面我们会真正写完整版，现在先给你看最终结构：

```
# CMPE 255 - Assignment 1
## End-to-End Data Science Using Generative AI

### Student
Haozheng Yang

## Overview

This assignment explores how a generative AI coding assistant
can support an end-to-end data science workflow.

The assignment contains two parts:

1. End-to-End Data Science Project
2. Replication of Data Science Prompt Experiments

---

# Part 1 - End-to-End Data Science

## Problem

...

## Dataset

...

## AI Assistant Used

ChatGPT

## Data Science Workflow

1. Data loading
2. Data cleaning
3. Exploratory data analysis
4. Feature engineering
5. Model training
6. Model evaluation
7. Model interpretation

## Results

...

## Interpretation

...

---

# Part 2 - Data Science Experiments
```

```
Experiments were replicated from the instructor repository.
```

```
| Experiment | Description |  
|---|---|  
| 1 | EDA |  
| 2 | Classification |  
| 3 | Hyperparameter Tuning |  
| 4 | Feature Importance |  
| ... | ... |
```

```
---
```

```
# Key Learnings
```

```
...
```

```
# YouTube Demo
```

```
[Watch the Assignment 1 walkthrough](YOUR_YOUTUBE_LINK)
```

之后我会帮你把这份 README 真正写成可以直接提交的英文版本，而不是这个 skeleton。

## YouTube 视频也不用你临场乱讲

我可以最终给你准备类似这样的录制流程：

```
0:00 Introduction
0:30 Assignment overview
1:00 GitHub repository structure
1:30 Part 1 problem statement
2:00 Dataset
2:30 AI prompts
3:30 Data cleaning
4:30 EDA
6:00 Model training
7:30 Model evaluation
9:00 Interpretation of results
10:30 Part 2 experiments
13:00 What I learned
14:00 Conclusion
```

以及逐段可以照着讲的英文稿。

**我们接下来最合适的动作是先完成 Part 1。**

你把老师那个 Google Doc **下载成 PDF/DOCX 上传到这里**，或者直接把里面的正文复制给我。我就可以准确读取老师给 Part 1 的数据集/任务，然后我们直接从：

**Step 1: 理解题目 → Step 2: 设计方案 → Step 3: 写 notebook/code → Step 4: 分析结果 → Step 5: 写 README → Step 6: YouTube script**

一路做到最后，不需要你自己猜下一步该做什么。

#4

**U** 你

没问题，我今天先完成part1，后面还有大约48小时可以完成part2，这是老师给的文档内容

没问题。现在有了老师的原始文档，我们可以把 Part 1 **今天完整收尾**，而且不需要把项目做得过度复杂。

我认真看了这份 5 页文档。老师对 Assignment 1 的核心要求其实非常明确：选择一个 **popular Kaggle dataset**，使用 ChatGPT 做一个 end-to-end data science 项目，最好涉及 deep learning，但简单的传统机器学习也完全可以；需要覆盖多个数据科学模块，并且提交完整的 chat transcript。

此外，还要求把实验中的部分内容整理成一篇 **Medium article/report**，并提交该文章。

更关键的是，老师在第一页直接说了：这些只是 **guidance**，如果做得比示例更好，可以获得额外分数。所以我们的目标不是复杂到难以完成，而是做一个**结构漂亮、流程完整、结果解释清楚、AI 使用过程明显**的项目。

## 今天 Part 1 的目标

我建议今天直接完成一个：

### *End-to-End Customer Churn Prediction using ChatGPT-Assisted Data Science*

这是比 Titanic 更合适的作业主题，因为它非常容易展示完整 Data Science workflow，而且最后有明确的 business interpretation。

我们可以覆盖：

```
Problem Definition
↓
Data Understanding
↓
Data Cleaning
↓
Exploratory Data Analysis
↓
Feature Engineering
↓
Preprocessing
↓
Model Training
↓
Model Comparison
↓
Model Evaluation
↓
Feature Importance
↓
Business Interpretation
↓
Conclusion
```

这实际上比老师示例要求的最低标准还稍微完整一点。

老师截图中的示范 prompt 也明显采用了 CRISP-DM 思路，包括 data understanding、data analysis、visualization、cleaning、preprocessing、feature selection、clustering/outlier analysis、regression/prediction、模型比较和最终推荐等步骤。我们不用机械复制，但可以采用相同的“阶段式实验”方式。□filecite□turn1file0□L19-L22□

## 我建议的数据集

我们用 Kaggle 上很经典的：

Telco Customer Churn

目标变量：

Churn

也就是说：

根据客户的账户信息、合同类型、使用服务、月费等信息，预测客户是否会流失。

它非常适合 classification。

例如可能出现这些变量：

```
gender
SeniorCitizen
Partner
Dependents
tenure
PhoneService
InternetService
OnlineSecurity
TechSupport
Contract
PaperlessBilling
PaymentMethod
MonthlyCharges
TotalCharges
Churn
```

最后可以回答非常容易理解的问题：

*Which customers are most likely to churn?*

以及：

*What factors are most strongly associated with customer churn?*

这在 YouTube 视频里也非常好讲。

## 模型不要只做一个

为了让项目显得完整，我们可以训练 3 个模型：

Logistic Regression  
Random Forest  
Gradient Boosting

然后建立这样的 comparison:

| Model               | Accuracy | Precision | Recall | F1  | ROC-AUC |
|---------------------|----------|-----------|--------|-----|---------|
| Logistic Regression | ...      | ...       | ...    | ... | ...     |
| Random Forest       | ...      | ...       | ...    | ... | ...     |
| Gradient Boosting   | ...      | ...       | ...    | ... | ...     |

最后解释哪个模型表现最好。

这样老师就能清楚看到:

ChatGPT 并不是只帮你跑了一个 sklearn model, 而是完成了一次真正的数据科学分析。

## 特别重要: 我们要留下 AI Assisted Data Science 的证据

因为老师明确要求:

*"using chatgpt"*

以及:

*"Export your chat transcript and submit it." □filecite□turn1file0□L25-L28□*

所以从现在开始, **这个 conversation 本身就是你的 assignment artifact之一**。

我们后面会把 workflow 刻意分阶段, 例如:

### Prompt 1 — Problem Understanding

你问我:

*Analyze this dataset and define the machine learning problem.*

然后得到分析。

### Prompt 2 — Data Understanding

*Perform dataset inspection and explain every important feature.*

### Prompt 3 — Data Cleaning

*Identify missing values, duplicates, incorrect data types, and potential data quality issues.*

### Prompt 4 — EDA

*Perform exploratory data analysis and generate meaningful visualizations.*

### Prompt 5 — Preprocessing

*Prepare the dataset for machine learning.*

### **Prompt 6 — Modeling**

*Train multiple classification models.*

### **Prompt 7 — Evaluation**

*Compare the models using appropriate classification metrics.*

### **Prompt 8 — Interpretation**

*Explain the results in simple language and identify important churn factors.*

这样以后导出聊天记录的时候，老师一打开就会看到我们是一步一步完成的。

---

## **Notebook 最终会长这样**

建议 notebook 名：

```
telco_customer_churn_end_to_end.ipynb
```

Notebook sections:

1. Project Overview
2. Import Libraries
3. Load Dataset
4. Data Understanding
  - shape
  - head()
  - info()
  - describe()
5. Data Quality Assessment
  - missing values
  - duplicates
  - data types
  - TotalCharges conversion
6. Exploratory Data Analysis
  - target distribution
  - churn by contract
  - churn by tenure
  - churn by internet service
  - monthly charges distribution
  - correlation analysis
7. Data Preprocessing
  - categorical encoding
  - numeric scaling
  - train/test split
8. Baseline Model
  - Logistic Regression
9. Machine Learning Models
  - Logistic Regression
  - Random Forest
  - Gradient Boosting
10. Model Evaluation
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - ROC-AUC
  - Confusion Matrix
11. Model Comparison
12. Feature Importance
13. Business Interpretation

这个已经完全覆盖老师所谓的“various modules of data science”。  
filecite turn1file0 L25-L28

### GitHub Part 1 今天最终应该长这样

```
Assignment_1/
├── README.md
├── Part_1/
│   ├── README.md
│   ├── telco_customer_churn_end_to_end.ipynb
│   ├── data/
│   │   └── WA_Fn-UseC_-Telco-Customer-Churn.csv
│   ├── images/
│   │   ├── churn_distribution.png
│   │   ├── churn_by_contract.png
│   │   ├── churn_by_tenure.png
│   │   ├── confusion_matrix.png
│   │   ├── model_comparison.png
│   │   └── feature_importance.png
│   ├── report/
│   │   └── medium_article.md
│   └── chat_transcript/
│       └── chatgpt_transcript.pdf
└── Part_2/
```

Part 2 今天先不碰。

### 今天的实际执行顺序

我们不要现在就写 README，因为**没有真实结果之前 README 只能是假数据**。

正确顺序是：

1. 下载数据集
2. 上传 CSV 给我
3. 我们一起分析数据
4. 做 Data Cleaning
5. 做 EDA
6. 做 visualization

7. preprocessing
  8. 训练模型
  9. 模型评估
  10. feature importance
  11. 我帮你逐图逐表写英文 interpretation
  12. 整理 .ipynb
  13. 写 GitHub README
  14. 写 Medium article
  15. 导出 ChatGPT transcript
  16. 准备 YouTube script
  17. 上传 GitHub
  18. 检查 GitHub public permission
- 所以你不用一次面对整个 assignment。

## 你现在只需要做一件事

去 Kaggle 搜索：

**Telco Customer Churn**

下载经典的 IBM Telco Customer Churn CSV，通常文件名类似：

```
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

然后**直接把 CSV 上传到这个聊天里**。

不要提前清洗，不要修改，也不用自己打开分析。

你把原始 CSV 发给我以后，我们就正式开始：

### *Part 1 — Step 1: Dataset Understanding & CRISP-DM Business Understanding*

从那一步开始，我会让整个过程既适合真正执行，又适合你最后直接拿去录 YouTube 和整理 GitHub。 □memcite□

#6

**U** 你

是在这里下载这个文件么

#7



CHATGPT

对，不过你现在打开的是别人做好的 Kaggle Notebook 页面，右上角这个 Download 菜单下载的是 notebook `.ipynb`，**不是我们需要的原始 CSV 数据集。**

你截图右侧有一个：

Input → Datasets → Telco Customer Churn

点击这个 Telco Customer Churn 数据集，然后进入数据集页面。

在那里找到 Download，我们需要下载的是数据集本身。下载后通常会得到 ZIP，解压后找：

```
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

或者名称类似的 `.csv` 文件。

**不要下载你截图里这个** `Download .ipynb`，因为那是其他人的分析代码。我们的目的正好是从原始 dataset 开始，让你和 ChatGPT 完成自己的 end-to-end analysis。

下载好 CSV 后直接拖到这里发给我，我们马上开始 Step 1: Business Understanding + Data Understanding。

#8



你

下载好了，是这份文件么

对，就是这份。✅

我已经实际读取检查过你上传的 CSV，不只是根据文件名判断。数据成功加载，基本信息是：

- 7,043 条客户记录
- 21 个 columns
- 目标变量是 `Churn`
- 包含 `tenure`、`Contract`、`InternetService`、`MonthlyCharges`、`TotalCharges`、`PaymentMethod` 等典型 Telco Customer Churn 特征
- `customerID` 是客户标识符
- 我还发现了一个之后需要处理的数据问题：`TotalCharges` 当前被读取成文本类型，而不是纯数值类型。这正好可以作为 Data Cleaning 的一部分。

所以数据完全正确，我们可以正式开始 Part 1。

而且建议从现在开始把后续对话当成你最后要提交的 ChatGPT-assisted data science transcript。我会按照老师要求的思路，一步一步进行，而不是一次性把整个答案生成完。

## Step 1 — Business Understanding

我们的项目可以定义为：

Project Title

*End-to-End Customer Churn Prediction Using AI-Assisted Data Science*

Business Problem

*A telecommunications company wants to understand why customers leave its service and identify customers who are at risk of churning. By analyzing customer demographics, subscribed services, contract information, tenure, and billing behavior, we aim to build a machine-learning model that predicts customer churn and identifies the major factors associated with customer attrition.*

Machine Learning Problem

这是一个 supervised binary classification problem：

```
Input: Customer information
      ↓
      ML Model
      ↓
Prediction: Churn = Yes / No
```

下一步就是正式进行 Step 2 — Data Understanding。在那里我们会真正检查这  $7,043 \times 21$  的 dataset，包括每个 column 的意义、数据类型、缺失值、target distribution 和基本统计信息，然后产生第一批可以放进 notebook/README 的真实结果。

#10



你

这些内容我是否需要编辑进入一个doc文本

不需要现在额外建一个 Word/Google Doc 来记录这些内容。

对于这个作业，我建议你把内容分成 3 个不同载体，不要重复劳动：

1. **ChatGPT 对话记录**：就是我们现在这整个聊天。老师明确要求 export chat transcript，所以我们一步步做实验，最后把完整聊天导出成 PDF 即可。老师原文要求 “Export your chat transcript and submit it.” □filecite□turn1file0□L23-L28□

1. **Jupyter/Colab Notebook**：这是最重要的实验 artifact。里面放代码、运行结果、图表，以及每一步的 Markdown 解释。例如：

```
## 1. Business Understanding
文字解释

## 2. Data Understanding
文字解释
↓
Python Code
↓
Output

## 3. Data Cleaning
文字解释
↓
Python Code
↓
Output

## 4. Exploratory Data Analysis
...
```

所以我刚才给你的 Business Problem / Machine Learning Problem，最终应该写进 `.ipynb` 的 Markdown cell，而不是单独 Word 文档。

1. **Medium Article**：等整个实验完成以后，我们再把 notebook 中最精彩的结果、图表和解释重新组织成一篇漂亮的文章。老师明确要求 publish a Medium article/report。□filecite□turn1file0□L29-L32□

最终就是：

```
我们的 ChatGPT 对话
↓
实际做实验
↓
.ipynb Notebook ← 代码 + 输出 + 解释
↓
实验全部完成
↓
README.md + Medium Article
↓
YouTube Walkthrough
↓
GitHub
```

所以**现在先不要花时间编辑 Doc**。

接下来最实际的一步是：**创建 Google Colab Notebook**。我们从第一格开始，我会告诉你每一个 Markdown cell 写什么、每一个 Code cell 写什么、运行以后应该观察什么，并且根据你的**真实运行结果**继续下一步。这样最终 notebook 本身就是完整作品，而不是事后拼出来的报告。

#12

U

你

他需要订阅50美金/月，这是必须的么

#13



CHATGPT

不，我不认为你现在需要为了这个 Assignment 去订阅 **\$50/月** 的方案。

老师文档第一页确实写了建议准备一些 subscription，其中包括 **Google Colab Pro subscription**，同时还列出了 ChatGPT/Gemini、GitHub Copilot 或其他 coding assistant 等。但老师在作业真正的 Part 1 要求里写的是：使用 ChatGPT，选择 Kaggle popular dataset，完成 data science 的多个模块并导出 chat transcript；并没有写“必须购买 \$50 Colab 套餐才能获得分数”。

尤其我们现在做的是 **Telco Customer Churn**，只有 **7,043 条数据**。我们计划的 Logistic Regression、Random Forest、Gradient Boosting 这些模型，对计算资源要求非常低，**完全没有必要为了它购买高档 GPU/Colab 方案**。

所以今天建议：

**先用免费的 Google Colab + 你现在的 ChatGPT 完成整个 Part 1。**

如果免费 Colab 真遇到资源限制，我们再处理。但对于这个 dataset，正常情况下 CPU 就足够。

另外，你看到的 **约 \$50/月** 很可能是 Colab 的更高级档位，而不应该把它理解成“Assignment 1 必须支付的费用”。老师文档更像是在列整个 semester 建议使用的工具环境，因为后面还涉及 Copilot、VS Code 等其他任务。

**所以先别花这 \$50。**

我们现在可以直接进入 Google Colab 免费版，创建一个：

```
CMPE255_Assignment1_Telco_Churn.ipynb
```

然后我从**第一个** Markdown cell开始带你做。

#14



你

我已准备好，我们开始吧

很好，你现在这个 Colab 已经准备好了。我们从 **最第一格** 开始，并且按“老师能看懂、最后能直接交”的方式来做。

先不要急着写代码。第一格建议做成 Markdown/Text cell，作为项目标题和背景说明。点击上方 + Text，然后粘贴以下内容：

```
# CMPE 255 Assignment 1
## End-to-End Customer Churn Prediction Using AI-Assisted Data Science

### Project Overview
In this project, I use ChatGPT as an AI-assisted data science tool to perform an end-to-end machine learning workflow on the Telco Customer Churn dataset.

The main goal is to predict whether a telecommunications customer will churn based on customer demographics, subscribed services, account information, contract type, tenure, and billing behavior.

This project follows a structured data science workflow inspired by CRISP-DM:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Exploratory Data Analysis
5. Feature Engineering and Preprocessing
6. Machine Learning Modeling
7. Model Evaluation and Comparison
8. Model Interpretation
9. Business Insights and Conclusion

### Machine Learning Problem
This is a supervised binary classification problem.

**Target variable:** `Churn`

- `Yes` = the customer left the company
- `No` = the customer remained with the company

### Objective
The objectives of this analysis are to:

- Understand the structure and quality of the Telco Customer Churn dataset.
- Identify patterns associated with customer churn.
- Build multiple machine learning classification models.
- Compare the models using appropriate evaluation metrics.
- Identify important features associated with customer churn.
- Translate the model results into meaningful business insights.
```

这一格**没有代码，不需要** Run Python。它就是 notebook 的 introduction。

接下来点击 + Code，创建第二格。

## Code Cell 1 — Import Libraries

粘贴：

```
# Import libraries for data manipulation
import pandas as pd
import numpy as np

# Import libraries for visualization
import matplotlib.pyplot as plt

# Display settings
pd.set_option('display.max_columns', None)

print("Libraries imported successfully!")
```

运行以后，你应该看到：

```
Libraries imported successfully!
```

这里我暂时只让你导入最基础的包。**我们不会一开始把几十个 library 全塞进去**，后面需要机器学习模型时再导入 sklearn。这样 notebook 更容易理解，也比较像正常学生一步一步完成的项目。

然后我们需要把 CSV 上传到 Colab。

你截图左侧有一个 **文件夹**  **图标**。

点击它 → 点击 Upload → 上传：

```
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

上传完成以后，在左侧应该能看到文件。

然后添加第三个 Code cell：

## Code Cell 2 — Load Dataset

```
# Load the Telco Customer Churn dataset
file_path = "/content/WA_Fn-UseC_-Telco-Customer-Churn.csv"

df = pd.read_csv(file_path)

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)
```

运行。

正常情况下应该看到：

```
Dataset loaded successfully!
Dataset shape: (7043, 21)
```

如果出现 (7043, 21)，就说明数据读取完全成功。

然后添加一个 Text cell:

```
## 1. Data Understanding
```

```
The first stage of the analysis is to understand the structure of the dataset.  
I will inspect the number of rows and columns, preview several customer records, examine  
data types, and review basic descriptive statistics.
```

```
This step helps identify potential data quality issues before performing visualization or  
machine learning.
```

下面创建新的 Code cell。

### Code Cell 3 — Preview the Dataset

```
# Display the first five rows of the dataset  
df.head()
```

运行以后你会看到前 5 个 customer records。

接着再创建一个 Code cell:

### Code Cell 4 — Dataset Dimensions

```
# Check the number of rows and columns  
rows, columns = df.shape  
  
print("Number of rows:", rows)  
print("Number of columns:", columns)
```

预计:

```
Number of rows: 7043  
Number of columns: 21
```

再创建一个 Code cell:

### Code Cell 5 — Column Names

```
# Display all column names  
print("Columns in the dataset:")  
  
for column in df.columns:  
    print(column)
```

这一步是刻意写得比较直观，而不是用复杂的 Python tricks。

然后:

### Code Cell 6 — Data Types

```
# Check column data types and non-null values  
df.info()
```

这里你要特别注意：

```
TotalCharges
```

你很可能会看到：

```
TotalCharges    object
```

这就是我们的第一个 Data Quality 问题。

理论上 `TotalCharges` 应该是数字，但目前 Pandas 把它识别成了文字。

**先不要修。**

Data Science 正确流程应该是：

先发现问题 → 解释问题 → 然后在 *Data Cleaning* 阶段处理。

这样你的 notebook 才有故事线。

最后再运行：

### Code Cell 7 — Descriptive Statistics

```
# Generate descriptive statistics for numerical columns
df.describe()
```

这里通常会看到：

```
SeniorCitizen
tenure
MonthlyCharges
```

等 numerical columns 的：

```
count
mean
std
min
25%
50%
75%
max
```

---

### 这一轮你先做到这里

所以现在你的 Colab 大约应该是：

Title / Project Overview

↓

Import Libraries

↓

Load Dataset

↓

1. Data Understanding

↓

`df.head()`

↓

Dataset Dimensions

↓

Column Names

↓

`df.info()`

↓

`df.describe()`

**先不要继续 Data Cleaning。**

尤其是 `df.info()` 和 `df.describe()`，我们下一步需要根据你实际运行出来的数据进行解释，而不是我提前假设结果。

你全部运行完成以后，**截一张包含 `df.info()` 输出和 `df.describe()` 输出的截图给我**；如果一张截不下，可以发两张。

然后我们进入：

### Step 2 — Interpret the Dataset + Data Quality Assessment

在那里我会告诉你 notebook 的 Markdown 应该怎么解释这些真实结果，并开始检查：

Missing Values  
Duplicates  
Incorrect Data Types  
Unique Values  
Target Distribution

这一步会正式发现 `TotalCharges` 的问题，然后再进入 Data Cleaning。□memcite□

#16

U 你

很好，这一步完全正确，而且你截图里的结果已经让我们发现了一个很好的 **真实 Data Quality Issue**。

从你的实际输出可以看到：

- Dataset: 7,043 rows × 21 columns
- SeniorCitizen : int64
- tenure : int64
- MonthlyCharges : float64
- TotalCharges : object ← 有问题
- Churn : object
- 所有列在 `df.info()` 中暂时都显示 7043 non-null

另外 `df.describe()` 给出的几个很适合写进报告的结果是：

- 平均 tenure ≈ 32.37 months
- tenure 范围：0–72 months
- 平均 MonthlyCharges ≈ \$64.76
- MonthlyCharges 范围：\$18.25–\$118.75

现在我们进入 **Step 2 — Data Quality Assessment**。

### 先添加一个 Text Cell

放在 `df.describe()` 下面：

```
```markdown id="0exm9k"
```

#### Initial Observations

The dataset contains 7,043 customer records and 21 columns.

From the initial inspection:

- Most customer-related categorical variables are stored as `object`.
- `tenure` and `SeniorCitizen` are stored as integers.
- `MonthlyCharges` is correctly stored as a numerical variable.
- `TotalCharges` is stored as an `object`, although it represents a monetary value and should normally be numerical. This suggests that the column may contain non-numeric or blank values that require further investigation.
- The target variable `Churn` is categorical and contains the customer churn outcome.

The descriptive statistics also show that the average customer tenure is approximately 32.37 months, while the average monthly charge is approximately \$64.76.

Before performing exploratory data analysis and machine learning, the dataset should be checked for missing values, duplicate records, and invalid values.

这里非常重要的一点是我们\*\*没有直接说 `TotalCharges` 一定有 missing values\*\*。

我们说：

> “may contain non-numeric or blank values that require further investigation.”

这才符合 Data Science 流程，因为现在我们只是从 `object` dtype \*\*怀疑存在问题\*\*。

---

## 接下来：检查 Missing Values

添加 Code cell:

```
```python id="wpp6pn"
# Check for missing values
missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)
```

你运行之后大概率会看到全部：

```
```text id="f9u1fe"
```

0

**\*\*不要觉得奇怪。\*\***

因为 `TotalCharges` 的问题很可能不是 Python 的 `NaN`，而是类似 `""` 的空字符串。

这正好给我们的 notebook 一个很好的数据清洗案例。

---

## 检查重复数据

再创建：

```
```python id="4bnl0n"
# Check for duplicate rows
duplicate_count = df.duplicated().sum()

print("Number of duplicate rows:", duplicate_count)
```

看看结果。

---

## 检查 categorical columns 的 unique values

然后创建：

```
```python id="w90fk8"
```

## Examine unique values in categorical columns

```
categorical_columns = df.selectdtypes(include='object').columns
for column in categorical_columns:
    print("\nColumn:", column)
    print(df[column].unique())
```

这里输出会比较长。

我们主要想看看：

```
```text id="lhv7aa"
gender
Partner
Dependents
PhoneService
MultipleLines
InternetService
OnlineSecurity
OnlineBackup
DeviceProtection
TechSupport
StreamingTV
StreamingMovies
Contract
PaperlessBilling
PaymentMethod
Churn
```

有没有奇怪值。

`customerID` 每个人基本都不同，所以它会输出很多 ID，影响可读性。

因此实际上我更推荐你把刚才代码**改进成下面这样**：

```
```python id="vm6d0k"
```

## Examine unique values in categorical columns

### Exclude customerID because it is only an identifier

```
categorical_columns = df.selectdtypes(include='object').columns
for column in categorical_columns:
    if column != 'customerID':
        print("\nColumn:", column)
        print(df[column].unique())
```

这样 notebook 会漂亮很多。

---

## 最重要的一步：调查 TotalCharges

创建 Code cell:

```
```python id="dqvl3"
# Investigate the TotalCharges column
print("Data type of TotalCharges:", df['TotalCharges'].dtype)

# Convert TotalCharges to numeric temporarily
# Invalid values will become NaN
total_charges_numeric = pd.to_numeric(df['TotalCharges'], errors='coerce')

print("Invalid or missing TotalCharges values:",
      total_charges_numeric.isnull().sum())
```

这一格非常关键。

`errors='coerce'` 的意思是：

如果某个 `TotalCharges` 无法转换成数字，就把它变成 `NaN`。

因此我们就能找出到底有多少 problematic values。

## 再进一步看看这些客户是谁

再添加：

```
```python id="ztu36q"
```

## Display rows where TotalCharges cannot be converted to a number

```
invalidtotalcharges = df[totalchargesnumeric.isnull()]

invalidtotalcharges[
['customerID', 'tenure', 'MonthlyCharges', 'TotalCharges', 'Churn']
]
```

这里我们是在做真正的 **Data Quality Investigation**，而不是发现 object 后马上粗暴删除。

如果结果显示这些客户：

```
```text id="3o3n4a"
tenure = 0
TotalCharges = blank
```

那就非常有意思。

我们就可以推断：

这些可能是刚加入公司的新客户，因此还没有累计 *Total Charges*。

但现在先不要做这个结论，等真实 output 出来。

## 最后检查 Target Distribution

再创建：

```
```python id="xrd7i2"
```

## Examine the target variable

```
print("Churn counts:")
print(df['Churn'].value_counts())

print("\nChurn percentages:")
print(df['Churn'].value_counts(normalize=True) * 100)
```

这个结果非常重要，因为我们需要判断：

> Dataset 是否存在 `class imbalance`。

如果例如出现：

```
```text id="lvcw3k"
No      ~73%
Yes     ~27%
```

那我们后面就不能只看 Accuracy。

因为一个永远预测 `No` 的笨模型，也能拿到大约 73% accuracy。

所以之后我们才会合理地使用：

**Precision + Recall + F1 + ROC-AUC + Confusion Matrix**

而不是为了让 notebook 看起来复杂随便堆 metrics。

## 这一轮你需要运行 5 个检查

```
```text id="e0j6qk"
```

1. Missing Values

↓

1. Duplicate Rows

↓

1. Categorical Unique Values

↓

1. TotalCharges Investigation

↓

## 1. Churn Distribution

然后把 **``TotalCharges Investigation` + `Churn Distribution`** 的结果截图发给我**``**。

下一步我们就进入非常关键的：

> **``Step 3 – Data Cleaning``**

届时我们会根据真实结果决定到底应该 **``删除、填补还是保留``** **``TotalCharges``** 的异常记录，而不是提前拍脑袋决定。之后马上开始 EDA 和第一批图表。

#18

U

你

很好，这个结果非常干净，而且现在我们已经有了足够证据进入 Step 3 — Data Cleaning。

从你的真实结果可以得出两个重要发现。

第一，`TotalCharges` 中确实有 11 条无法转换成数值的记录。更关键的是，你截图中的这 11 位客户全部满足：

```
tenure = 0
TotalCharges = blank
```

这说明它并不像随机缺失，更可能是这些客户刚刚加入，还没有产生累计费用。因此我们有两个合理选择：把 `TotalCharges` 填成 0，或者删除这 11 条记录。这里我建议填成 0，因为 `tenure = 0` 与尚未产生累计费用在业务逻辑上是一致的，而且没必要丢掉这些真实客户。

第二，Churn distribution 是：

```
No = 5174 (73.46%)
Yes = 1869 (26.54%)
```

所以存在明显的 class imbalance。之后模型评估不能只看 accuracy，还应该重点看 Recall、Precision、F1 和 ROC-AUC。

## Step 3 — Data Cleaning

先在刚才的结果下面添加一个 Text Cell：

```
## 2. Data Cleaning
```

```
The initial data quality assessment identified an important issue in the `TotalCharges` column.
```

```
Although `TotalCharges` represents a monetary value, it was stored as an `object` data type. Further investigation showed that 11 records contained blank values that could not be converted directly to numbers.
```

```
All 11 of these customers have a `tenure` value of 0, indicating that they are new customers who have not yet accumulated total charges. Instead of removing these customers, the blank `TotalCharges` values will be replaced with 0. This preserves valid customer records while maintaining a logical interpretation of the data.
```

```
The target variable is also moderately imbalanced:
```

```
- No Churn: 5,174 customers (73.46%)
- Churn: 1,869 customers (26.54%)
```

```
Because of this imbalance, model performance will later be evaluated using multiple metrics rather than accuracy alone, including precision, recall, F1-score, ROC-AUC, and the confusion matrix.
```

这一段以后可以直接用于 README 和 Medium。

## Code Cell — Convert TotalCharges

下面创建 Code cell:

```
# Convert TotalCharges from object to numeric
# Invalid blank values are converted to NaN
df['TotalCharges'] = pd.to_numeric(
    df['TotalCharges'],
    errors='coerce'
)

# Fill missing TotalCharges values with 0
# These records correspond to new customers with tenure = 0
df['TotalCharges'] = df['TotalCharges'].fillna(0)

print("TotalCharges data type:", df['TotalCharges'].dtype)
print("Missing TotalCharges values:", df['TotalCharges'].isnull().sum())
```

正常应该得到类似:

```
TotalCharges data type: float64
Missing TotalCharges values: 0
```

## Code Cell — Check duplicates

如果你前面已经运行过 duplicate 检查, 而且结果是 0, 就**不要重复运行**。

如果还没运行:

```
# Check for duplicate rows
duplicate_count = df.duplicated().sum()

print("Number of duplicate rows:", duplicate_count)
```

如果是 0, 很好, 不需要删除任何记录。

## Code Cell — Check customer IDs

我们还要检查 customerID 是否真的每个客户唯一:

```
# Check whether customerID is unique
unique_customers = df['customerID'].nunique()

print("Number of rows:", len(df))
print("Number of unique customer IDs:", unique_customers)
```

理想结果:

```
Number of rows: 7043
Number of unique customer IDs: 7043
```

这说明:

*Each row represents one unique customer.*

同时也告诉我们，`customerID` 后面**不应该作为** ML feature，因为它只是 identifier。

## Cleaning 后做一次最终检查

创建：

```
# Final data quality check after cleaning
print("Dataset shape:", df.shape)
print("\nMissing values:")
print(df.isnull().sum())

print("\nData types:")
print(df.dtypes)
```

我们希望最后还是：

```
Dataset shape: (7043, 21)
```

并且：

```
TotalCharges    float64
```

所有 missing values：

```
0
```

这样 Data Cleaning 就正式结束。

## 接下来不要马上 Machine Learning

下一阶段是这次作业视觉效果最好、也是 Medium 和 YouTube 最容易展示的部分：

## Step 4 — Exploratory Data Analysis (EDA)

我们会开始生成真正可以保存到 GitHub 的图：

```
Churn Distribution
↓
Churn by Gender
↓
Churn by Senior Citizen
↓
Churn by Contract Type
↓
Churn by Internet Service
↓
Churn by Payment Method
↓
Tenure vs Churn
↓
Monthly Charges vs Churn
↓
Total Charges vs Churn
↓
Correlation Analysis
```

不过我们不会为了数量机械画 10 张图。我会挑出真正有意义的几张，并且**每画一张就解释它告诉我们什么**。

这会让最终 notebook 呈现出：

*Question → Visualization → Observation → Business Interpretation*

而不是普通学生作业里常见的“连续放十张图但什么都不解释”。

你现在先运行上面的 cleaning。**完成以后，把最后那个** `Final data quality check` **的输出截图发给我。**

确认数据已经 clean 后，我们正式开始 EDA **第一组图表**。

#20

U

你

这个报错很简单，不是你的 Data Cleaning 代码有问题。截图里真正的错误是：

```
NameError: name 'pd' is not defined
```

意思是当前 Colab runtime 里 Pandas 的缩写 `pd` 不存在了。

最常见的原因是 Colab runtime 中途断开/重启了。你之前虽然运行过：

```
import pandas as pd
```

但 runtime 一旦重启，之前存在内存里的 `pd`、`df` 等变量都会消失。

### 最推荐的解决方法

不要修改 Cleaning 代码。

直接点击顶部：

Runtime → Run all

或者你截图顶部的 Run all。

这样 Colab 会按照顺序重新执行：

```
import pandas as pd
↓
读取 CSV
↓
df 创建
↓
前面的 Data Understanding
↓
TotalCharges Cleaning
```

但是有一个问题：Colab runtime 重启后，你上传到 `/content/` 的 CSV 也可能消失。

所以先看看左侧 📁 文件夹里还有没有：

```
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

如果没有，就重新上传一次。

然后再 Run all。

**为什么不建议只在这里加** `import pandas as pd`

虽然你也可以直接：

```
import pandas as pd
```

然后继续，但很可能下一步又出现：

```
NameError: name 'df' is not defined
```

因为 `df` 也可能随着 runtime 重启消失。

所以最安全的是从 notebook 顶部重新运行。

另外，这其实是 Colab 很正常的现象：

*Notebook 保存了代码和输出，但 runtime 中的 Python variables 并不会永久保存。*

所以以后重新打开 notebook 时，通常就是：

**上传/挂载 dataset → Run all**

你现在先确认 CSV 还在 → Run all。如果全部成功，刚才这段 Cleaning 最后应该输出：

```
TotalCharges data type: float64  
Missing TotalCharges values: 0
```

然后继续运行我们后面的 `customerID` 和 final quality check。

把最终输出给我，我们接着做 EDA。

#22



你

对，这次报错已经很明确了，而且说明刚才的判断是对的：Colab runtime 重启后，你之前上传的 CSV 消失了。

截图里的核心错误是：

```
FileNotFoundError:
No such file or directory:
'/content/WA_Fn-UseC_-Telco-Customer-Churn.csv'
```

代码本身没问题。

### 现在这样操作

点击 Colab 左侧的  文件夹图标，然后选择 Upload，重新上传：

```
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

上传完成后，你应该能在左边看到类似：

```
sample_data
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

然后再点击：

Runtime → Run all

或者顶部 Run all。

这样整个 notebook 会从：

```
import pandas as pd
```

开始重新运行，然后重新：

```
df = pd.read_csv(file_path)
```

之后前面的 Data Understanding、Data Quality Assessment、Cleaning 都会重新执行。

### 一个 Colab 使用习惯

以后如果你关闭 Colab、断开 runtime，或者第二天重新打开，代码还在，但 `/content/` 里手动上传的 CSV 通常不会永久保留。

所以每次新 session：

重新上传 CSV → Run all

就可以恢复整个实验环境。

我们之后也可以把 notebook 改成从 Google Drive 读取 dataset，这样不用每次重新上传；但今天为了尽快完成 Part 1，不需要折腾这个。

你现在重新上传 CSV → Run all。如果后面又出现红色报错，直接截图给我；如果全部绿色通过，就把最后的 Final Data Quality Check 发给我，我们继续 EDA。

#24 U 你

对，现在上传成功了。👍

从截图看，左侧 Files 最下面已经出现了 CSV：

```
WA_Fn-UseC_-Telco-Custome...
```

所以现在**不需要再上传**。

不过我注意到一个关键点：你现在文件似乎位于 Colab 当前文件系统的根目录视图，而我们代码写的是：

```
file_path = "/content/WA_Fn-UseC_-Telco-Customer-Churn.csv"
```

为了避免再次猜路径，最简单的办法是：在左侧这个 CSV 文件上点击右侧的：（鼠标移动到文件名附近通常会出现），选择：

**Copy path**

然后把复制出来的路径粘贴给我，或者直接把它替换到：

```
file_path = "这里放复制的路径"
```

例如如果 Copy path 得到：

```
/WA_Fn-UseC_-Telco-Customer-Churn.csv
```

那就改成：

```
file_path = "/WA_Fn-UseC_-Telco-Customer-Churn.csv"
```

如果得到：

```
/content/WA_Fn-UseC_-Telco-Customer-Churn.csv
```

就保持原来的代码。

**不要手动猜路径，直接 Copy path 最稳。**

然后重新运行这个 Load Dataset cell。只要看到：

```
Dataset loaded successfully!  
Dataset shape: (7043, 21)
```

就说明恢复成功。

之后再点 Runtime → Run all。如果全部运行成功，我们就继续 EDA。

是的，我暂离了一段时间，现在我放到正确的文件夹内了，我们可以继续了

可以，我们继续。既然 CSV 已经放回正确位置，先确认 Run all 能顺利跑到我们刚才的 Data Cleaning 结束。如果没有红色报错，我们现在正式进入 Step 4 — Exploratory Data Analysis (EDA)。

这一阶段我们会开始制作之后 README、Medium 和 YouTube 都能使用的图表。

## Step 4 — EDA

在 Cleaning 后面添加一个 Text Cell:

```
## 3. Exploratory Data Analysis
```

```
After cleaning the dataset, exploratory data analysis (EDA) is performed to better understand customer behavior and identify patterns associated with churn.
```

```
The analysis focuses on several important questions:
```

1. How common is customer churn?
2. Is churn associated with contract type?
3. How does customer tenure relate to churn?
4. Are monthly charges different between churned and retained customers?
5. Does internet service type affect churn behavior?
6. Which other customer characteristics appear to be associated with churn?

```
These observations will help guide feature preprocessing and machine learning model development.
```

## EDA 1 — Churn Distribution

先做最重要的 target visualization:

```
# Calculate churn counts
churn_counts = df['Churn'].value_counts()

# Create bar chart
plt.figure(figsize=(7, 5))
plt.bar(churn_counts.index, churn_counts.values)

plt.title('Customer Churn Distribution')
plt.xlabel('Churn')
plt.ylabel('Number of Customers')

plt.show()

# Display exact values
print(churn_counts)
print("\nChurn Percentage:")
print(df['Churn'].value_counts(normalize=True) * 100)
```

运行后应该与你之前的数据一致:

```
No    5174
Yes    1869
```

也就是大约:

```
No    73.46%
Yes    26.54%
```

图出来后, 在下面添加一个 Text Cell:

```
### Observation: Churn Distribution
```

```
The dataset contains considerably more non-churned customers than churned customers.
Approximately 73.46% of customers remained with the company, while 26.54% churned.
```

```
This indicates a moderate class imbalance. Therefore, accuracy alone may not be
sufficient for evaluating the machine learning models. Metrics such as precision, recall,
F1-score, ROC-AUC, and the confusion matrix will also be considered.
```

这个 interpretation 是根据你前面实际运行出来的结果写的, 不是猜测。

## EDA 2 — Churn by Contract Type

下面做一个非常有 business value 的分析。

```
# Create a cross-tabulation of Contract and Churn
contract_churn = pd.crosstab(
    df['Contract'],
    df['Churn']
)

print(contract_churn)

# Visualize churn by contract type
contract_churn.plot(
    kind='bar',
    figsize=(8, 5)
)

plt.title('Customer Churn by Contract Type')
plt.xlabel('Contract Type')
plt.ylabel('Number of Customers')
plt.xticks(rotation=0)
plt.legend(title='Churn')
plt.show()
```

这张图运行后先不要写 interpretation。

因为我们要根据你实际看到的数字来解释。

## EDA 3 — Churn Rate by Contract Type

上一张图展示数量，但数量有一个问题：

假设 Month-to-month 本来客户就特别多，它的 churn 数量自然可能比较高。

所以我们再计算 churn rate：

```
# Calculate churn rate for each contract type
contract_churn_rate = pd.crosstab(
    df['Contract'],
    df['Churn'],
    normalize='index'
) * 100

print(contract_churn_rate)
```

这会告诉我们类似：

|                | No  | Yes |
|----------------|-----|-----|
| Contract       |     |     |
| Month-to-month | ... | ... |
| One year       | ... | ... |
| Two year       | ... | ... |

这里的 Yes 才是：

在这种 contract 的客户中，有百分之多少最终 churn？

这个分析比单纯 count 更有意义。

---

## EDA 4 — Tenure vs Churn

接下来看看客户待得越久是否越不容易离开：

```
# Compare customer tenure between churn groups
plt.figure(figsize=(8, 5))

df[df['Churn'] == 'No']['tenure'].hist(
    bins=20,
    alpha=0.6,
    label='No'
)

df[df['Churn'] == 'Yes']['tenure'].hist(
    bins=20,
    alpha=0.6,
    label='Yes'
)

plt.title('Customer Tenure Distribution by Churn')
plt.xlabel('Tenure (Months)')
plt.ylabel('Number of Customers')
plt.legend(title='Churn')

plt.show()

# Compare average tenure
print("Average tenure by churn status:")
print(df.groupby('Churn')['tenure'].mean())
```

这里我特意用了比较基础的 Pandas + Matplotlib，没有引入复杂绘图库。

## EDA 5 — Monthly Charges vs Churn

```
# Compare MonthlyCharges between churn groups
plt.figure(figsize=(7, 5))

df.boxplot(
    column='MonthlyCharges',
    by='Churn'
)

plt.title('Monthly Charges by Churn Status')
plt.suptitle('')
plt.xlabel('Churn')
plt.ylabel('Monthly Charges ($)')

plt.show()

# Display average monthly charges
print("Average Monthly Charges by churn status:")
print(df.groupby('Churn')['MonthlyCharges'].mean())
```

这里会产生一个 boxplot。

我们想回答：

## 这一轮先停在这里

现在你的 notebook 应该开始有明显的 Data Science 分析感觉了：

### 3. Exploratory Data Analysis

Churn Distribution



Interpretation

Churn by Contract



Contract Churn Rate

Tenure vs Churn



Average Tenure

Monthly Charges vs Churn



Average Monthly Charges

先把 EDA 2-5 全部运行。

然后给我两张截图就够了：

1. Contract churn rate 的数字 + 图
2. Tenure 和 MonthlyCharges 的输出/图

我会根据你的真实结果给你写下一组 Markdown Observations & Business Interpretation。

之后我们再补 2-3 个真正有价值的 categorical analyses，然后 EDA 就结束。不需要为了显得复杂画二十张图。下一阶段就是 Preprocessing → Machine Learning。□memcite□

#28

U 你

很好，这三张图已经出现了非常清楚的故事，而且比单纯跑模型更有价值。我们先把**真实结果解释**写进 Notebook，然后再补最后两组 EDA。

## 先写 EDA Interpretation

在这三组分析下面新增一个 Text Cell，粘贴：

### ### EDA Findings and Interpretation

#### #### 1. Contract Type and Customer Churn

Contract type shows a strong relationship with customer churn.

The dataset contains:

- Month-to-month contracts: 2,220 retained customers and 1,655 churned customers.
- One-year contracts: 1,307 retained customers and 166 churned customers.
- Two-year contracts: 1,647 retained customers and only 48 churned customers.

Customers with month-to-month contracts appear much more likely to churn, while customers with longer-term contracts show substantially lower churn.

This suggests that contract commitment may be an important factor in customer retention. Long-term contracts could provide greater customer stability, while month-to-month customers represent a higher-risk group for churn.

#### #### 2. Customer Tenure and Churn

Customer tenure also shows a clear relationship with churn.

The average tenure is:

- No churn: approximately 37.57 months
- Churn: approximately 17.98 months

Customers who churn tend to have much shorter relationships with the company. The tenure distribution also shows a particularly high number of churned customers among customers with relatively low tenure.

This suggests that the early stage of the customer relationship may be especially important for retention. Newer customers may require additional engagement or retention strategies.

#### #### 3. Monthly Charges and Churn

Customers who churn also tend to have higher monthly charges.

The average monthly charge is:

- No churn: approximately \$61.27
- Churn: approximately \$74.44

The boxplot similarly shows that the distribution of monthly charges for churned customers is shifted toward higher values.

This suggests that higher monthly costs may be associated with increased churn risk. However, this relationship should not be interpreted as causal because other factors, such as service type and contract type, may also influence both monthly charges and churn.

最后一句 “should not be interpreted as causal” 很值得保留。

这是比较好的 Data Science interpretation:

*我们发现 association, 但不能因为一张图就说高价格“导致” churn。*

## Contract Churn Rate 还缺一个输出

你第一张截图给了 raw counts:

|                |         |          |
|----------------|---------|----------|
| Month-to-month | No 2220 | Yes 1655 |
| One year       | No 1307 | Yes 166  |
| Two year       | No 1647 | Yes 48   |

但我之前让你算的这个:

```
contract_churn_rate = pd.crosstab(
    df['Contract'],
    df['Churn'],
    normalize='index'
) * 100

print(contract_churn_rate)
```

截图里没有显示。

运行一下。

根据你现在的 counts, 我计算出来应该大约是:

| Churn<br>Contract | No     | Yes    |
|-------------------|--------|--------|
| Month-to-month    | 57.29% | 42.71% |
| One year          | 88.73% | 11.27% |
| Two year          | 97.17% | 2.83%  |

但 notebook 里还是让 Python 自己产生真实结果。

这组数字特别适合 Medium:

*42.7% of month-to-month customers churned, compared with only 2.8% of customers on two-year contracts.*

这个比“1655 vs 48”更有分析意义。

## EDA 6 — Internet Service

接下来分析 service type。

新增 Code Cell:

```
# Analyze churn by Internet Service type
internet_churn_rate = pd.crosstab(
    df['InternetService'],
    df['Churn'],
    normalize='index'
) * 100

print("Churn rate by Internet Service:")
print(internet_churn_rate)

# Visualize the churn rate
internet_churn_rate['Yes'].plot(
    kind='bar',
    figsize=(8, 5)
)

plt.title('Churn Rate by Internet Service')
plt.xlabel('Internet Service')
plt.ylabel('Churn Rate (%)')
plt.xticks(rotation=0)

plt.show()
```

这里我们不再画 No/Yes count。

因为现在已经理解了：

*Rate 往往比 raw count 更适合比较不同大小的 customer groups。*

看看：

```
DSL
Fiber optic
No
```

哪个 churn rate 最高。

## EDA 7 — Payment Method

新增：

```
# Analyze churn rate by Payment Method
payment_churn_rate = pd.crosstab(
    df['PaymentMethod'],
    df['Churn'],
    normalize='index'
) * 100

print("Churn rate by Payment Method:")
print(payment_churn_rate)

# Visualize churn rate
payment_churn_rate['Yes'].sort_values(ascending=False).plot(
    kind='bar',
    figsize=(9, 5)
)

plt.title('Churn Rate by Payment Method')
plt.xlabel('Payment Method')
plt.ylabel('Churn Rate (%)')
plt.xticks(rotation=30, ha='right')

plt.show()
```

这个也很可能会出现明显差异。

先不要写 interpretation，等真实结果。

## 再加一个稍微高级一点的分析：Correlation

因为我们的 notebook 想到要比最低要求稍好一些，可以加入 numerical feature correlation。

先创建：

```
# Select numerical variables for correlation analysis
numerical_data = df[
    [
        'SeniorCitizen',
        'tenure',
        'MonthlyCharges',
        'TotalCharges'
    ]
]

correlation_matrix = numerical_data.corr()

print("Correlation Matrix:")
print(correlation_matrix)
```

然后画：

```
# Visualize the correlation matrix
plt.figure(figsize=(7, 5))

plt.imshow(
    correlation_matrix,
    cmap='coolwarm',
    aspect='auto'
)

plt.colorbar(label='Correlation')

plt.xticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns,
    rotation=45
)

plt.yticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns
)

plt.title('Correlation Matrix of Numerical Features')

plt.show()
```

这里你会开始看到例如：

`tenure` 和 `TotalCharges` 是否有明显 correlation？

这是很合理的，因为：

```
TotalCharges
≈
MonthlyCharges × Time
```

所以 `tenure` 越长，累计费用通常也越高。

不过同样：

*correlation ≠ causation*

## 一个小地方顺便修掉

你第三张截图顶部出现了：

```
<Figure size 700x500 with 0 Axes>
```

这是因为之前代码：

```
plt.figure(figsize=(7, 5))

df.boxplot(...)
```

`df.boxplot()` 自己创建了 figure, 所以前面的 `plt.figure()` 变成了一张空 figure。

不影响结果, 但是 GitHub Notebook 里看起来不够漂亮。

把 Monthly Charges 那个 cell 改成:

```
# Compare MonthlyCharges between churn groups
df.boxplot(
    column='MonthlyCharges',
    by='Churn',
    figsize=(7, 5)
)

plt.title('Monthly Charges by Churn Status')
plt.suptitle('')
plt.xlabel('Churn')
plt.ylabel('Monthly Charges ($)')

plt.show()

# Display average monthly charges
print("Average Monthly Charges by churn status:")
print(df.groupby('Churn')['MonthlyCharges'].mean())
```

重新运行, 就不会出现那个:

```
<Figure size 700x500 with 0 Axes>
```

了。

---

## 做完这三个, 我们就停止 EDA

现在不要无限画图。

最终我们已经覆盖:

```
Churn Distribution
  ↓
Contract Type ★
  ↓
Tenure ★
  ↓
Monthly Charges ★
  ↓
Internet Service
  ↓
Payment Method
  ↓
Numerical Correlation
```

其中 ★ 是目前最明显的三个发现。

你下一次给我 Internet Service + Payment Method + Correlation Matrix 的结果即可。

然后我帮你写最终 EDA Summary，我们就正式进入：

#### Step 5 — Machine Learning Preprocessing

包括：

X/y → 删除 customerID → train/test split → categorical encoding → numerical scaling → Logistic Regression

之后开始真正训练模型。

#30

U 你

是的，你推测的没错，这是之前运算的结果，以及后续的

对，这组结果很好，而且已经足够让我们**结束 EDA 阶段**了。现在数据故事已经非常清楚，不需要继续堆图。

从你的真实输出看，有几个很强的发现：

- **Contract**: Month-to-month churn rate **42.71%**, One year **11.27%**, Two year 只有 **2.83%**。
- **Internet Service**: Fiber optic churn **41.89%**, DSL **18.96%**, No internet **7.40%**。
- **Payment Method**: Electronic check churn **45.29%**, 明显高于 Mailed check **19.11%**、Bank transfer automatic **16.71%**、Credit card automatic **15.24%**。
- **Correlation**: `tenure` 与 `TotalCharges` 的 Pearson correlation  $\approx 0.826$ ; `MonthlyCharges` 与 `TotalCharges`  $\approx 0.651$ 。这在业务上也合理，因为累计费用自然会受到客户使用时间和月费影响。

结合前面的结果，我们现在已经形成一个完整的 EDA story：

*Churn is especially concentrated among newer customers, month-to-month customers, customers with higher monthly charges, fiber-optic customers, and electronic-check users.*

但注意我们说的是 **associated with churn**，不是说这些因素一定“cause churn”。

## 先把 EDA Summary 写进去

在 correlation 图下面添加一个 Text Cell：

```
```markdown id="j16ahv"
```

### EDA Summary and Business Insights

The exploratory data analysis revealed several important patterns associated with customer churn.

#### Contract Type

Contract type shows one of the strongest relationships with churn. The churn rate for month-to-month customers is approximately 42.71%, compared with 11.27% for one-year contracts and only 2.83% for two-year contracts. This suggests that customers with longer-term commitments are substantially less likely to leave the company.

#### Customer Tenure

Customers who churn have a much shorter average tenure. Churned customers have an average tenure of approximately 17.98 months, compared with 37.57 months for retained customers. This suggests that the earlier stages of the customer relationship may represent an important period for customer retention.

#### Monthly Charges

Churned customers pay higher monthly charges on average. Their average monthly charge is approximately \$74.44, compared with \$61.27 among customers who remain with the company.

#### Internet Service

Fiber optic customers have the highest churn rate at approximately 41.89%. This is considerably higher than DSL customers at approximately 18.96% and customers without internet service at

approximately 7.40%.

### **Payment Method**

Customers using electronic checks have the highest churn rate at approximately 45.29%. In comparison, customers using automatic bank transfers or automatic credit card payments have churn rates of approximately 16.71% and 15.24%, respectively.

### **Numerical Correlations**

The correlation analysis shows a strong positive relationship between `tenure` and `TotalCharges` (approximately 0.83). `MonthlyCharges` and `TotalCharges` also show a moderately strong positive correlation (approximately 0.65). These relationships are reasonable because total accumulated charges are influenced by both the duration of the customer relationship and monthly service costs.

### **Overall EDA Conclusion**

The analysis suggests that customers with month-to-month contracts, shorter tenure, higher monthly charges, fiber optic internet service, and electronic check payments represent potentially higher-risk customer groups.

These findings describe associations in the dataset and should not automatically be interpreted as causal relationships. The next stage will use machine learning to determine whether customer churn can be predicted from these characteristics.

这个可以作为你之后 Medium article 最重要的一部分。

---

## # Step 5 – Data Preprocessing

现在正式进入 Machine Learning。

先添加：

```
```markdown id="i3m0kv"
```

### ## 4. Data Preprocessing for Machine Learning

Before training machine learning models, the dataset must be transformed into a format suitable for classification algorithms.

The preprocessing process includes:

1. Separating the target variable (`Churn`) from the input features.
2. Removing `customerID` because it is a unique identifier rather than a meaningful predictive feature.
3. Converting the target variable into binary values.
4. Separating numerical and categorical features.
5. Encoding categorical variables using one-hot encoding.
6. Scaling numerical features where appropriate.
7. Splitting the dataset into training and testing sets.

To reduce data leakage, preprocessing transformations will be learned from the training data through a machine learning pipeline rather than being applied to the entire dataset before the train-test split.

最后这句话很重要。它说明我们知道 data leakage 是什么。

## 1. 导入 sklearn

Code Cell:

```
```python id="sfm6ts"
```

### Import preprocessing and machine learning tools

```
from sklearn.modelselection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
```

### Import machine learning models

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
```

## Import evaluation metrics

```
from sklearn.metrics import (
    accuracyscore,
    precisionscore,
    recallscore,
    f1score,
    rocaucscore,
    confusionmatrix,
    classificationreport
)

print("Machine learning libraries imported successfully!")
```

```
---

# 2. 创建 X 和 y

```python id="y9njz2"
# Separate input features and target variable

# customerID is removed because it is only a unique identifier
X = df.drop(columns=['customerID', 'Churn'])

# Convert target variable:
# No = 0, Yes = 1
y = df['Churn'].map({
    'No': 0,
    'Yes': 1
})

print("Feature matrix shape:", X.shape)
print("Target shape:", y.shape)

print("\nTarget distribution:")
print(y.value_counts())
```

预期:

```
```text id="0bv3q3"
Feature matrix shape: (7043, 19)
Target shape: (7043,)
```

以及:

```
```text id="7j6yip"
0    5174
1    1869
```

### 3. Numerical / Categorical Features

```
```python id="2u2yhe"
```

#### Identify numerical and categorical features

```
numerical_features = X.selectdtypes(
include=['int64', 'float64']
).columns.tolist()

categorical_features = X.selectdtypes(
include=['object']
).columns.tolist()

print("Numerical features:")
print(numerical_features)

print("\nCategorical features:")
print(categorical_features)

print("\nNumber of numerical features:", len(numerical_features))
print("Number of categorical features:", len(categorical_features))
```

这里你应该能看到：

```
```text id="bnrm5k"
Numerical:
SeniorCitizen
tenure
MonthlyCharges
TotalCharges
```

剩下的是 categorical。

---

### 4. Train/Test Split

```
```python id="a4icm8"
```

#### Split the dataset into training and testing sets

```
Xtrain, Xtest, ytrain, ytest = train_test_split(
X,
y,
testsize=0.20,
randomstate=42,
stratify=y
)

print("Training feature shape:", Xtrain.shape)
print("Testing feature shape:", Xtest.shape)
```

```
print("\nTraining target distribution:")
print(ytrain.valuecounts(normalize=True) * 100)
print("\nTesting target distribution:")
print(ytest.valuecounts(normalize=True) * 100)
```

这里用了：

```
```python id="31jt91"
stratify=y
```

原因非常重要。

我们的：

```
```text id="kpk10a"
```

No  $\approx$  73.46%

Yes  $\approx$  26.54%

存在 class imbalance。

`stratify=y` 会让 training/test set 尽量保持相同的 churn proportion。

---

# 5. 建立 Preprocessor

```
```python id="txc3x8"
# Create preprocessing pipelines

# Numerical features will be standardized
numerical_transformer = Pipeline(
    steps=[
        ('scaler', StandardScaler())
    ]
)

# Categorical features will be converted using one-hot encoding
categorical_transformer = Pipeline(
    steps=[
        ('onehot', OneHotEncoder(
            handle_unknown='ignore'
        ))
    ]
)

# Combine numerical and categorical preprocessing
preprocessor = ColumnTransformer(
    transformers=[
        (
            'num',
            numerical_transformer,
            numerical_features
        ),
        (
            'cat',
            categorical_transformer,
            categorical_features
        )
    ]
)

print("Preprocessing pipeline created successfully!")
```

这里暂时**不要** `fit_transform(X)`。

因为下一步我们会把：

```
```text id="1fqdkx"
```

Preprocessor

+

Machine Learning Model

组成一个完整的 Pipeline。

这意味着：

```
```text id="6dzv5a"
Training Data
    ↓
Fit preprocessing
    ↓
Transform features
    ↓
Train model
```

而 Test Data 只会：

```
```text id="pbph70"
Test Data
    ↓
Use already learned preprocessing
    ↓
Prediction
```

这样就避免 test data 偷偷参与训练。

---

# Step 6 – 第一个真正的 ML Model

我们先做 **Logistic Regression baseline**。

添加 Text：

```
```markdown id="zpmz2q"
## 5. Machine Learning Modeling

### Model 1: Logistic Regression

Logistic Regression is used as the first baseline classification model. It provides a relatively simple and interpretable starting point for predicting customer churn.

The preprocessing pipeline and classifier are combined into a single machine learning pipeline to ensure that preprocessing is learned only from the training data.
```

然后：

```
```python id="q2j4hp"
```

## Create Logistic Regression pipeline

```
logisticmodel = Pipeline(
steps=[
```

```
('preprocessor', preprocessor),  
('classifier', LogisticRegression(  
    maxiter=1000,  
    random_state=42  
))  
]  
)
```

## Train the model

```
logisticmodel.fit(Xtrain, y_train)  
  
print("Logistic Regression training completed!")
```

然后预测：

```
```python id="3qxx32"  
# Make predictions using the test dataset  
y_pred_logistic = logistic_model.predict(X_test)  
  
# Obtain probabilities for ROC-AUC  
y_prob_logistic = logistic_model.predict_proba(X_test)[ :, 1]  
  
print("Predictions completed!")
```

---

## 最精彩的部分：Evaluation

运行：

```
```python id="pryj4a"
```

### Evaluate Logistic Regression

```
logisticaccuracy = accuracy_score(  
    ytest,  
    ypred_logistic  
)  
  
logisticprecision = precision_score(  
    ytest,  
    ypred_logistic  
)  
  
logisticrecall = recall_score(  
    ytest,  
    ypred_logistic  
)  
  
logisticf1 = f1_score(  
    ytest,
```

```

ypred_logistic
)

logisticrocauc = rocaucscore(
ytest,
yprob_logistic
)

print("Logistic Regression Performance")
print("-----")
print(f"Accuracy: {logisticaccuracy:.4f}")
print(f"Precision: {logisticprecision:.4f}")
print(f"Recall: {logisticrecall:.4f}")
print(f"F1 Score: {logisticf1:.4f}")
print(f"ROC-AUC: {logisticrocauc:.4f}")

```

然后：

```

```python id="zysip7"
# Display detailed classification report
print(
    classification_report(
        y_test,
        y_pred_logistic,
        target_names=['No Churn', 'Churn']
    )
)

```

## Confusion Matrix

最后：

```

```python id="gjxshm"

```

## Create confusion matrix

```

cm = confusionmatrix(
ytest,
ypredlogistic
)

print("Confusion Matrix:")
print(cm)

plt.figure(figsize=(6, 5))

plt.imshow(
cm,
cmap='Blues'
)

```

```
plt.title('Logistic Regression Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.xticks([0, 1], ['No Churn', 'Churn'])
plt.yticks([0, 1], ['No Churn', 'Churn'])
```

## Add values inside the matrix

```
for i in range(2):
    for j in range(2):
        plt.text(
            j,
            i,
            cm[i, j],
            ha='center',
            va='center'
        )
plt.colorbar()
plt.show()
```

---

### 这一轮先做到 Logistic Regression

暂时\*\*不要跑 Random Forest 和 Gradient Boosting\*\*。

因为现在会得到五个非常重要的真实数字：

```
```text id="yxxzqy"
Accuracy
Precision
Recall
F1
ROC-AUC
```

以及：

```
```text id="k4xbxi"
TN FP
FN TP
```

我们需要先理解 `baseline model` 到底做得怎么样。

特别是 **Recall**:

> 在真正会 `churn` 的客户中，我们成功抓到了多少？

对于 `Customer Churn`，这往往比单纯 `Accuracy` 更值得关注。

所以你运行到 **Confusion Matrix** 后，把：

**Logistic Regression Performance + Classification Report + Confusion Matrix**

截图发给我。

下一步我会先带你**真正读懂这些指标**，然后训练：

> **Random Forest → Gradient Boosting → Model Comparison → ROC Curve → Feature Importance  
→ Final Business Recommendation**

到那里 `Machine Learning` 主体基本就完成了。📄[memcite](#)📄

#32



你

很好，这个 Logistic Regression baseline 跑得完全正常，而且结果非常适合这个 assignment。现在不要急着只说“80.55% accuracy 很好”，我们需要像 data scientist 一样解释。

## 1. Logistic Regression 的真实结果

你的模型得到：

| Metric    | Result | 含义                                 |
|-----------|--------|------------------------------------|
| Accuracy  | 80.55% | 所有客户中约 80.55% 被正确分类                |
| Precision | 65.72% | 被模型预测为 churn 的客户中约 65.72% 真的 churn |
| Recall    | 55.88% | 真正 churn 的客户中约 55.88% 被模型发现        |
| F1 Score  | 60.40% | Precision 与 Recall 的综合表现           |
| ROC-AUC   | 84.21% | 模型区分 churn / no-churn 客户的整体能力较好    |

特别值得注意的是：

*Accuracy 80.55% 看起来不错，但 churn recall 只有 55.88%。*

这正好证明了我们之前为什么不能只看 Accuracy。

## 2. Confusion Matrix 怎么理解

你的结果：

```
[[926 109]
 [165 209]]
```

对应：

|            | Predicted |     |
|------------|-----------|-----|
|            | No        | Yes |
| Actual No  | 926       | 109 |
| Actual Yes | 165       | 209 |

也就是说：

926 True Negatives

*926 个实际上没有 churn 的客户被正确预测为 No Churn。*

109 False Positives

*109 个实际上没有 churn 的客户被错误预测成 Churn。*

165 False Negatives

⚠️ 165 个实际上 churn 的客户，模型却预测他们不会 churn。

## 209 True Positives

209 个真正 churn 的客户被模型成功识别出来。

从 customer retention 的角度，False Negative 特别值得关注。

因为：

公司以为这个客户不会走，所以可能不会采取 retention action，但实际上他最终 churn 了。

这就是为什么 Recall = 55.88% 很重要。

## 把 Interpretation 写进 Notebook

在 Logistic Regression 结果下面添加 Text Cell：

```
```markdown id="c7j2fn"
```

### Logistic Regression Results and Interpretation

The Logistic Regression baseline model achieved the following performance:

- **Accuracy:** 80.55%
- **Precision:** 65.72%
- **Recall:** 55.88%
- **F1-score:** 60.40%
- **ROC-AUC:** 84.21%

The overall accuracy indicates that the model correctly classified approximately 81% of customers in the test dataset. However, because the dataset contains more non-churn customers than churn customers, accuracy alone does not provide a complete evaluation of model performance.

The ROC-AUC score of approximately 0.84 indicates that the model has a good overall ability to distinguish between customers who churn and customers who remain.

### Confusion Matrix Interpretation

The confusion matrix produced the following results:

- **True Negatives:** 926 customers were correctly predicted as non-churn.
- **False Positives:** 109 customers were predicted to churn but actually remained.
- **False Negatives:** 165 customers were predicted to remain but actually churned.
- **True Positives:** 209 customers were correctly identified as churned customers.

The churn recall is approximately 55.88%, meaning that the model successfully identifies slightly more than half of the customers who actually churn.

From a customer retention perspective, false negatives are particularly important because these customers are at risk of leaving but would not be identified by the model for potential retention actions.

Therefore, while Logistic Regression provides a strong baseline with good overall discrimination, there is still room to improve the identification of churned customers.

这段以后非常适合你在 YouTube 里 paraphrase, 不需要逐字念。

---

# Step 7 – Random Forest

现在看看一个 tree-based model 能不能超过 Logistic Regression。

添加 Text Cell:

```
```markdown id="t9aq1l"
```

```
### Model 2: Random Forest
```

Random Forest is an ensemble learning algorithm that combines multiple decision trees. Unlike Logistic Regression, it can capture more complex and nonlinear relationships between customer characteristics and churn.

The model will be trained using the same training and testing datasets to allow a fair comparison with the Logistic Regression baseline.

Code:

```
```python id="gy3ezr"
```

## Create Random Forest pipeline

```
randomforestmodel = Pipeline(  
steps=[  
( 'preprocessor', preprocessor),  
( 'classifier', RandomForestClassifier(  
n_estimators=200,  
randomstate=42  
))  
]  
)
```

## Train Random Forest

```
randomforestmodel.fit(Xtrain, ytrain)  
print("Random Forest training completed!")
```

预测:

```
```python id="6cecrw"
# Make Random Forest predictions
y_pred_rf = random_forest_model.predict(X_test)

# Obtain churn probabilities
y_prob_rf = random_forest_model.predict_proba(X_test)[: , 1]

print("Random Forest predictions completed!")
```

Evaluation:

```
```python id="4j2krk"
```

## Evaluate Random Forest

```
rfaccuracy = accuracyscore(ytest, ypredrf)
rfprecision = precisionscore(ytest, ypredrf)
rfrecall = recallscore(ytest, ypredrf)
rff1 = f1score(ytest, ypredrf)
rfroauc = rocaucscore(ytest, yprob_rf)

print("Random Forest Performance")
print("-----")
print(f"Accuracy: {rfaccuracy:.4f}")
print(f"Precision: {rfprecision:.4f}")
print(f"Recall: {rfrecall:.4f}")
print(f"F1 Score: {rff1:.4f}")
print(f"ROC-AUC: {rfroauc:.4f}")
```

先不用画它的 `confusion matrix`。最终最佳模型再重点画。

---

# Step 8 – Gradient Boosting

添加 Text Cell:

```
```markdown id="34b2fx"
### Model 3: Gradient Boosting

Gradient Boosting is another ensemble learning method that builds decision trees sequentially, with each new tree attempting to correct errors made by previous trees.

This provides a third modeling approach and allows the performance of linear and tree-based classification methods to be compared.
```

Code:

```
```python id="0q0e6s"
```

## Create Gradient Boosting pipeline

```
gradientboostingmodel = Pipeline(  
steps=[  
( 'preprocessor', preprocessor),  
( 'classifier', GradientBoostingClassifier(  
random_state=42  
))  
]  
)
```

## Train Gradient Boosting

```
gradientboostingmodel.fit(Xtrain, ytrain)  
  
print("Gradient Boosting training completed!")
```

预测:

```
```python id="f7azc1"  
# Make Gradient Boosting predictions  
y_pred_gb = gradient_boosting_model.predict(X_test)  
  
# Obtain churn probabilities  
y_prob_gb = gradient_boosting_model.predict_proba(X_test)[:, 1]  
  
print("Gradient Boosting predictions completed!")
```

Evaluation:

```
```python id="sqj41b"
```

## Evaluate Gradient Boosting

```
gbaccuracy = accuracyscore(ytest, ypredgb)  
gbprecision = precisionscore(ytest, ypredgb)  
gbrecall = recallscore(ytest, ypredgb)  
gbf1 = f1score(ytest, ypredgb)  
gbroauc = rocaucscore(ytest, yprob_gb)  
  
print("Gradient Boosting Performance")  
print("-----")  
print(f"Accuracy: {gbaccuracy:.4f}")  
print(f"Precision: {gbprecision:.4f}")  
print(f"Recall: {gbrecall:.4f}")  
print(f"F1 Score: {gbf1:.4f}")  
print(f"ROC-AUC: {gbroauc:.4f}")
```

```
---
```

### # Step 9 – 三个模型正面对比

这一步会成为你最终报告里非常重要的一张表。

```
```python id="dyxkyc"  
# Create model comparison table
```

```
model_results = pd.DataFrame({  
    'Model': [  
        'Logistic Regression',  
        'Random Forest',  
        'Gradient Boosting'  
    ],  
  
    'Accuracy': [  
        logistic_accuracy,  
        rf_accuracy,  
        gb_accuracy  
    ],  
  
    'Precision': [  
        logistic_precision,  
        rf_precision,  
        gb_precision  
    ],  
  
    'Recall': [  
        logistic_recall,  
        rf_recall,  
        gb_recall  
    ],  
  
    'F1 Score': [  
        logistic_f1,  
        rf_f1,  
        gb_f1  
    ],  
  
    'ROC-AUC': [  
        logistic_roc_auc,  
        rf_roc_auc,  
        gb_roc_auc  
    ]  
})  
  
model_results.round(4)
```

你会得到类似：

| Model               | Accuracy | Precision | Recall | F1    | ROC-AUC |
|---------------------|----------|-----------|--------|-------|---------|
| Logistic Regression | .8055    | .6572     | .5588  | .6040 | .8421   |
| Random Forest       | ?        | ?         | ?      | ?     | ?       |
| Gradient Boosting   | ?        | ?         | ?      | ?     | ?       |

现在**不要预设** Random Forest **一定最好**。  
很可能结果会让你意外。

## 再画 Model Comparison

```
```python id="onuw5"
```

### Compare model performance visually

```
comparisonplot = modelresults.set_index('Model')[
    ['Accuracy', 'Precision', 'Recall', 'F1 Score', 'ROC-AUC']
]

comparison_plot.plot(
    kind='bar',
    figsize=(11, 6)
)

plt.title('Machine Learning Model Performance Comparison')
plt.ylabel('Score')
plt.xlabel('Model')
plt.ylim(0, 1)
plt.xticks(rotation=0)
plt.legend(loc='lower right')
plt.show()
```

这张图之后非常适合：

- GitHub README
- Medium
- YouTube

---

## 为什么我们现在不直接宣布“最佳模型”

因为这里有一个非常重要的 Data Science 问题：

假设结果变成：

```
```text
Model A:
Accuracy = 82%
Recall = 50%

Model B:
Accuracy = 80%
Recall = 65%
```

哪个更好？

没有单一答案。

如果业务目标是：

*Find customers likely to churn so the company can intervene*

那么 Recall 可能比一点点 Accuracy 更重要。

如果 retention campaign 成本非常高，Precision 又可能更加重要。

这正是老师希望看到的 “read and understand and provide a paraphrase of the output”——不是把 AI/代码生成的数字念一遍，而是说明这些数字意味着什么。

你现在把 Random Forest + Gradient Boosting + Model Comparison Table 跑完。

下一条把 comparison table 截图发给我。

然后我们完成 Part 1 数据科学主体最后几个非常漂亮的步骤：

Best Model Selection → ROC Curves → Feature Importance → Final Business Recommendations  
→ Final Conclusion

之后就不继续增加 ML 复杂度了，我们转向 GitHub README + Medium Article + YouTube walkthrough + Chat transcript。

#34



你

很好，这个结果其实比“三个模型中某个模型全面碾压”更适合作业，因为它让我们有真正的 model-selection reasoning 可以讲。

你的结果是：

| Model               | Accuracy | Precision | Recall | F1     | ROC-AUC |
|---------------------|----------|-----------|--------|--------|---------|
| Logistic Regression | 0.8055   | 0.6572    | 0.5588 | 0.6040 | 0.8421  |
| Random Forest       | 0.7835   | 0.6186    | 0.4813 | 0.5414 | 0.8206  |
| Gradient Boosting   | 0.8027   | 0.6655    | 0.5160 | 0.5813 | 0.8434  |

这里不存在“所有指标都最好的模型”。

**Logistic Regression** 有最高 Accuracy、Recall 和 F1；**Gradient Boosting** 有最高 Precision 和 ROC-AUC；Random Forest 在当前配置下整体落后。

对于我们的业务目标——**尽可能识别潜在 churn customers**——我建议最终选择 **Logistic Regression 作为主要模型**。因为它的 churn Recall 是三个模型最高的 55.88%，F1 也是最高的，而且 ROC-AUC 只比 Gradient Boosting 低 0.0013。

## 先写 Model Comparison Interpretation

在 comparison chart 后添加 Text Cell：

### ### Model Comparison and Selection

Three classification models were evaluated using the same training and testing datasets: Logistic Regression, Random Forest, and Gradient Boosting.

The results show that no single model dominates every evaluation metric.

- **Logistic Regression** achieved the highest accuracy (80.55%), recall (55.88%), and F1-score (60.40%).
- **Gradient Boosting** achieved the highest precision (66.55%) and ROC-AUC (84.34%).
- **Random Forest** produced lower performance across most metrics in the current experiment.

Gradient Boosting achieved the highest ROC-AUC, although the difference compared with Logistic Regression was very small (0.8434 vs. 0.8421).

For this customer churn problem, recall is particularly important because a false negative represents a customer who actually churns but is not identified as being at risk. Missing these customers may prevent the company from applying retention strategies.

Therefore, Logistic Regression is selected as the primary model because it provides the highest recall and F1-score while maintaining strong accuracy and ROC-AUC performance.

The comparison also demonstrates that a more complex model does not necessarily produce better results. Model selection should depend on the business objective and appropriate evaluation metrics rather than model complexity alone.

最后一句特别值得你 YouTube 里讲。

---

## Step 10 — ROC Curves

现在做最后一张 model comparison 图：ROC。

首先在 import 那里补：

```
from sklearn.metrics import roc_curve
```

然后运行：

```

# Calculate ROC curves for all three models
fpr_logistic, tpr_logistic, _ = roc_curve(
    y_test,
    y_prob_logistic
)

fpr_rf, tpr_rf, _ = roc_curve(
    y_test,
    y_prob_rf
)

fpr_gb, tpr_gb, _ = roc_curve(
    y_test,
    y_prob_gb
)

# Plot ROC curves
plt.figure(figsize=(8, 6))

plt.plot(
    fpr_logistic,
    tpr_logistic,
    label=f'Logistic Regression (AUC = {logistic_roc_auc:.3f})'
)

plt.plot(
    fpr_rf,
    tpr_rf,
    label=f'Random Forest (AUC = {rf_roc_auc:.3f})'
)

plt.plot(
    fpr_gb,
    tpr_gb,
    label=f'Gradient Boosting (AUC = {gb_roc_auc:.3f})'
)

# Random classifier reference line
plt.plot(
    [0, 1],
    [0, 1],
    linestyle='--',
    label='Random Classifier'
)

plt.title('ROC Curves for Customer Churn Models')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend()
plt.grid(alpha=0.3)

plt.show()

```

这张图的 interpretation 可以直接写：

### ### ROC Curve Interpretation

The ROC curves compare the ability of the three models to distinguish between churned and non-churned customers across different classification thresholds.

Gradient Boosting achieved the highest ROC-AUC (0.8434), followed very closely by Logistic Regression (0.8421). Random Forest achieved a lower ROC-AUC of 0.8206.

The very small difference between Gradient Boosting and Logistic Regression indicates that both models have similar overall discrimination ability. Logistic Regression remains the selected primary model because it achieved better recall and F1-score at the default classification threshold.

注意这里出现了一个很重要的新词：

*default classification threshold*

现在 `predict()` 通常相当于使用 0.5 左右的 decision threshold。

其实如果我们把 threshold 降到例如：

0.40

0.35

Recall 很可能会上升。

这可以成为我们超越最低要求的最后一个实验。

---

## Step 11 — Feature Importance / Interpretation

因为最终选 Logistic Regression，我们不要只看 Random Forest feature importance，而是解释 Logistic Regression coefficients。

运行：

```
# Get feature names after preprocessing
feature_names = logistic_model.named_steps[
    'preprocessor'
].get_feature_names_out()

# Get Logistic Regression coefficients
coefficients = logistic_model.named_steps[
    'classifier'
].coef_[0]

# Create coefficient DataFrame
feature_effects = pd.DataFrame({
    'Feature': feature_names,
    'Coefficient': coefficients
})

# Sort by coefficient
feature_effects = feature_effects.sort_values(
    by='Coefficient',
    ascending=False
)

print("Features most associated with higher predicted churn:")
display(feature_effects.head(10))

print("\nFeatures most associated with lower predicted churn:")
display(feature_effects.tail(10))
```

这里有一个解释规则：

```
Positive coefficient
    ↓
Associated with higher predicted churn

Negative coefficient
    ↓
Associated with lower predicted churn
```

但千万不要写：

*"This feature causes churn."*

仍然只能说：

*"associated with higher/lower predicted churn."*

## 做一张 Feature Coefficient 图

为了保持图表容易阅读，我们只取影响最明显的一部分：

```

# Select strongest positive and negative coefficients
top_positive = feature_effects.head(8)
top_negative = feature_effects.tail(8)

important_features = pd.concat([
    top_negative,
    top_positive
])

# Plot coefficients
plt.figure(figsize=(10, 7))

plt.barh(
    important_features['Feature'],
    important_features['Coefficient']
)

plt.axvline(
    x=0,
    linestyle='--'
)

plt.title('Important Features in Logistic Regression')
plt.xlabel('Logistic Regression Coefficient')
plt.ylabel('Feature')

plt.tight_layout()
plt.show()

```

这里运行之后，**先不要自己写** interpretation。

我要根据你的真实 coefficient 来判断哪些因素最重要。

## Step 12 — 我建议加一个 Threshold Experiment

这个实验非常值得做，因为它能让你的项目从普通：

*"I trained three models."*

提升成：

*"I considered the business cost of false negatives and investigated classification threshold selection."*

而且代码很简单。

先测试几个 threshold：

```
# Evaluate different classification thresholds
threshold_results = []

for threshold in [0.30, 0.35, 0.40, 0.45, 0.50]:

    predictions = (
        y_prob_logistic >= threshold
    ).astype(int)

    threshold_results.append({
        'Threshold': threshold,
        'Precision': precision_score(
            y_test,
            predictions
        ),
        'Recall': recall_score(
            y_test,
            predictions
        ),
        'F1 Score': f1_score(
            y_test,
            predictions
        )
    })

threshold_results_df = pd.DataFrame(
    threshold_results
)

threshold_results_df.round(4)
```

你应该会看到一个很有意思的 trade-off:

```
Threshold ↓
    ↓
更多客户被判断为 churn
    ↓
Recall ↑
Precision ↓
```

这正是 churn retention 的真实业务决策。

📄 `genui` {"learningviz":{"typeid":"CLASSIFICATIONTHRESHOLD","localeoverride":"zh-CN"}}📄

你可以用上面的交互图理解为什么 threshold 改变会同时改变 TP、FP、TN、FN，以及 Precision/Recall。

## 现在这一轮只做三个东西

不要继续增加内容了：

① ROC Curve

② Logistic Regression Feature Coefficients

### ③ Threshold Experiment Table

把 Feature Coefficients 的两个 table + Threshold Experiment Table 截图给我。

下一步我会根据你的真实结果写：

*Feature Interpretation → Threshold Recommendation → Business Recommendations → Final Conclusion*

到那一步，我们的 End-to-End Data Science 实验主体就正式完成。

之后今天剩下的工作就从“做 Data Science”切换成“包装提交成果”：整理 Notebook → GitHub README → Medium article → YouTube script → transcript。□memcite□

#36



你

非常好。到这里，Part 1 的 Data Science / Machine Learning 主体已经基本完成了。而且 threshold experiment 给了我们一个很好的“超出最低要求”的分析点。

你的 threshold 结果尤其有价值：

| Threshold | Precision     | Recall        | F1            |
|-----------|---------------|---------------|---------------|
| 0.30      | 0.5193        | <b>0.7540</b> | <b>0.6150</b> |
| 0.35      | 0.5432        | 0.7059        | 0.6140        |
| 0.40      | 0.5682        | 0.6684        | 0.6143        |
| 0.45      | 0.6021        | 0.6150        | 0.6085        |
| 0.50      | <b>0.6572</b> | 0.5588        | 0.6040        |

最有意思的是：默认 threshold 0.50 并不是唯一合理选择。降到 0.30 后，Recall 从 55.88% → 75.40%，F1 也从 60.40% → 61.50%，代价是 Precision 从 65.72% → 51.93%。

所以如果公司的目标是 retention screening——宁可多联系一些最终不会 churn 的客户，也不希望漏掉大量真正会 churn 的客户——0.30 是一个值得考虑的 operating threshold。

## 先把最后的分析写进 Notebook

在 Feature Importance 和 Threshold Experiment 后添加这个 Text Cell：

```
```markdown id="mk0mh7"
```

## 6. Model Interpretation and Business Insights

### Feature Interpretation

The Logistic Regression coefficients provide additional insight into the characteristics associated with predicted customer churn.

Several features are associated with a higher predicted probability of churn:

- Fiber optic internet service
- Month-to-month contracts
- Higher total charges
- Electronic check payments
- Lack of online security
- Lack of technical support
- Streaming TV and streaming movie services

Among these categorical features, fiber optic internet service and month-to-month contracts show particularly strong positive coefficients.

Several features are associated with a lower predicted probability of churn:

- Longer customer tenure

- Two-year contracts
- DSL internet service
- Customers without internet service
- Non-paperless billing

Customer tenure has the strongest negative coefficient in the model, suggesting that customers who have remained with the company for longer periods tend to have substantially lower predicted churn risk.

These relationships are associations learned by the model and should not be interpreted as proof that any individual feature directly causes customer churn.

### **Connection with Exploratory Data Analysis**

The model interpretation is broadly consistent with the earlier exploratory analysis.

For example:

- Month-to-month customers had a churn rate of approximately 42.71%, while two-year contract customers had a churn rate of only approximately 2.83%.
- Fiber optic customers had a churn rate of approximately 41.89%.
- Electronic check customers had a churn rate of approximately 45.29%.
- Customers who churned had substantially shorter average tenure than customers who remained.

The consistency between the exploratory analysis and model coefficients increases confidence that these variables contain meaningful information for identifying customer churn risk.

这里最后一段很不错，因为你不是把 EDA 和 ML 当成两个完全分离的东西，而是在做：

> EDA发现 pattern → ML 又发现类似 pattern。

---

# Threshold Experiment Interpretation

下面再加一个 Text Cell:

```
```markdown id="jhhwx"
```

```
### Classification Threshold Analysis
```

The default Logistic Regression classification threshold of 0.50 produced:

- Precision: 65.72%
- Recall: 55.88%
- F1-score: 60.40%

However, customer churn prediction is a business problem in which false negatives can be costly. A false negative represents a customer who actually churns but is not identified as being at risk.

To investigate this trade-off, several classification thresholds were evaluated.

Lowering the threshold increased recall:

- Threshold 0.50 → Recall 55.88%
- Threshold 0.45 → Recall 61.50%
- Threshold 0.40 → Recall 66.84%
- Threshold 0.35 → Recall 70.59%
- Threshold 0.30 → Recall 75.40%

At a threshold of 0.30, the model identifies approximately three-quarters of the customers who actually churn. The F1-score also slightly improves from 0.6040 at the default threshold to 0.6150.

The trade-off is lower precision. At the 0.30 threshold, precision decreases to approximately 51.93%, meaning that more non-churning customers will also be classified as high-risk.

### Business Interpretation

If the cost of contacting customers with retention offers is relatively low compared with the cost of losing a customer, a lower threshold such as 0.30 could be useful for identifying a broader group of at-risk customers.

If retention interventions are expensive, a higher threshold may be preferable because it provides higher precision.

Therefore, the classification threshold should ultimately be selected based on the

business costs of false positives and false negatives rather than using 0.50 automatically.

这已经是一个相当不错的 ML discussion。

## 有一个 Feature Importance 细节我们要严谨一点

你看到：

```
```text id="nhrgg7"
```

```
numtenure -1.257539
```

```
numMonthlyCharges -0.591863
```

```
num__TotalCharges 0.536253
```

不要直接比较：

> tenure -1.25 比 Fiber optic 0.63 大一倍，所以 tenure 一定重要一倍。

因为 categorical dummy variables 和 standardized numerical variables 的 coefficient interpretation 不完全一样。

所以我们保持：

> “strong association”

而不是做过度 quantitative interpretation。

这也是为什么刚才给你的 Markdown 没有说：

> tenure is exactly twice as important.

---

## # Step 13 – Final Business Recommendations

现在进入整个 notebook 的最后一部分。

添加 Text Cell:

```
```markdown id="7qfr5p"
```

### ## 7. Business Recommendations

Based on the exploratory analysis and machine learning results, several potential customer retention strategies can be considered.

#### ### 1. Focus on New Customers

Customers who churn have substantially shorter tenure than retained customers. The company should pay particular attention to the early stages of the customer relationship.

Potential actions include improved onboarding, proactive customer support, and early satisfaction monitoring.

#### ### 2. Encourage Longer-Term Contracts

Month-to-month customers show a substantially higher churn rate than customers with one-year or two-year contracts.

The company could consider incentives that encourage suitable customers to transition from month-to-month plans to longer-term contracts.

#### ### 3. Investigate Fiber Optic Customer Experience

Fiber optic customers show a relatively high churn rate and fiber optic service is positively associated with predicted churn in the Logistic Regression model.

This does not prove that fiber optic service causes churn. Further analysis should

investigate possible explanations such as pricing, service quality, customer expectations, or differences in customer characteristics.

#### ### 4. Review Electronic Check Customers

Customers using electronic checks show a higher churn rate than customers using other payment methods.

The company could investigate whether payment experience, customer characteristics, or other factors explain this association and potentially encourage convenient automatic payment options where appropriate.

#### ### 5. Target Retention Campaigns Using Churn Risk

The machine learning model can be used to identify customers with elevated predicted churn risk.

For broader retention screening, lowering the classification threshold from 0.50 to approximately 0.30 substantially increases recall. This allows the company to identify more customers who eventually churn, although it also increases the number of false-positive alerts.

The appropriate threshold should therefore depend on the cost of retention interventions compared with the cost of customer churn.

注意我们的 business recommendations 都比较谨慎。

没有写：

*Force everyone into two-year contracts.*

而是：

*investigate / consider / encourage suitable customers*

这是更专业的表达。

## Step 14 — Final Conclusion

最后一个 Text Cell：

```markdown id="7y9tcm"

### 8. Final Conclusion

This project demonstrated an end-to-end AI-assisted data science workflow for customer churn prediction using the Telco Customer Churn dataset.

The analysis included:

- Business and data understanding
- Data quality assessment and cleaning
- Exploratory data analysis

- Feature preprocessing
- Logistic Regression
- Random Forest
- Gradient Boosting
- Multi-metric model evaluation
- ROC analysis
- Feature interpretation
- Classification threshold analysis
- Business recommendations

Among the three models evaluated, Logistic Regression was selected as the primary model. It achieved:

- **Accuracy:** 80.55%
- **Precision:** 65.72%
- **Recall:** 55.88%
- **F1-score:** 60.40%
- **ROC-AUC:** 84.21%

Gradient Boosting achieved a slightly higher ROC-AUC of 84.34%, but Logistic Regression provided better recall and F1-score at the default classification threshold.

The exploratory analysis and model interpretation consistently identified several characteristics associated with customer churn, including shorter tenure, month-to-month contracts, fiber optic internet service, electronic check payments, and higher monthly charges.

The threshold experiment further demonstrated that model deployment decisions should consider business objectives. Lowering the Logistic Regression threshold from 0.50 to 0.30 increased churn recall from 55.88% to 75.40%, allowing the model to identify substantially more customers who eventually churn at the cost of lower precision.

Overall, the project demonstrates that successful data science involves more than maximizing a single model metric. Data quality, exploratory analysis, model comparison, interpretation, business objectives, and the trade-off between different types of prediction errors all influence the final decision.

---

# Notebook 主体现在完成

到这里，你实际上已经完成：

```
```text id="d0jqlx"
Business Understanding
    ✓
Data Understanding
    ✓
Data Quality
    ✓
Data Cleaning
    ✓
EDA
    ✓
Preprocessing
    ✓
Logistic Regression
    ✓
Random Forest
    ✓
Gradient Boosting
    ✓
Model Comparison
    ✓
Confusion Matrix
    ✓
ROC Curve
    ✓
Feature Interpretation
    ✓
Threshold Experiment
    ✓
Business Recommendations
    ✓
Conclusion
    ✓
```

这已经明显超过“简单训练一个 Kaggle classifier”的最低标准。

## 现在做一次最终 Notebook 清理

不要继续增加模型了。

从 notebook 最顶部开始快速往下检查一次：

- 删除所有红色 error output，比如之前 `pd is not defined` 和 `FileNotFoundError` 的失败记录。
- 确认没有 `<Figure size ... with 0 Axes>` 这种空图。
- Markdown heading 顺序保持合理。
- 不要留下重复运行造成的重复图。

- 确认最终 `TotalCharges` 是 numerical。
- 确认所有图正常显示。
- 确认最后 Model Comparison 和 ROC Curve 都在。
- 确认最后有 Business Recommendations + Final Conclusion。

然后：

**Runtime → Restart session and run all**

重新上传 CSV（如果 Colab 要求）后，从头到尾 Run all 一次。

这个步骤非常重要。

因为它验证：

**整个 notebook 可以从原始 dataset 开始 reproducibly 跑到最终 conclusion。**

如果从头到尾没有 error，我们就把这个版本作为最终 notebook。

---

## 然后下载两个文件

Colab：

**File → Download → Download .ipynb**

得到：

```text id="jtmxb3"

CMPE255Assignment1Telco\_Churn.ipynb

同时建议：

**\*\*File → Print → Save as PDF\*\***

或者后面用 Colab/Jupyter 的其他方式保存一个 notebook PDF。

CSV 也保留。

---

# 下一阶段不再做 Data Science

现在我们正式进入 **\*\*Part 1 Submission Packaging\*\***:

```
```text id="djdak5"
Completed Notebook
    ↓
GitHub Repository
    ↓
Professional README.md
    ↓
Medium Article
    ↓
ChatGPT Transcript
    ↓
YouTube Walkthrough
    ↓
Final Submission
```

老师原始要求明确要求你使用 ChatGPT 在 Kaggle popular dataset 上完成 data science workflow 并导出 chat transcript。□filecite□turn1file0□L23-L28□ 同时还要求整理并提交 Medium article/report。□filecite□turn1file0□L29-L32□

**所以现在先完成上面最后几个 Markdown，并从头 Run all。**

如果全部通过，你下一条直接告诉我：

***“全部运行成功，没有 error。”***

然后我们马上开始制作 GitHub repository + README。这部分我可以直接给你生成完整的、基于你这些真实实验结果的 README，不需要你自己重新整理数字。□memcite□

#38

U

你

全部运行成功没有error，以及这是文件

很好，我也实际检查了你上传的 `CMPE255_Assignment1_Telco_Churn.ipynb`，不是只看文件名。

目前 notebook 状态很好：

- 共 68 个 cells
- 所有主要章节都存在，从 `Data Understanding` 一直到 `Final Conclusion`
- Notebook 保存了运行 outputs
- 我检查了 code-cell outputs，**没有保存任何 error output**
- Model Comparison、ROC、Feature Interpretation、Threshold Analysis、Business Recommendations 都已经包含
- 最终 Conclusion 也已经写入

所以这份 `.ipynb` 可以作为 **Part 1 的正式 artifact**。

接下来我建议我们按这个顺序完成提交包装：

GitHub repository → README.md → Medium article → Chat transcript → YouTube video/script → **最终检查**

其中现在最优先的是 **GitHub + README**，因为老师最终要求所有 artifacts 整理在 GitHub，而且 README 相当于整个 Assignment 1 的入口页面。

我建议 repository 组织成：

```
CMPE255-Assignment-1/
├── README.md
├── Part_1/
│   ├── CMPE255_Assignment1_Telco_Churn.ipynb
│   ├── WA_Fn-UseC_-Telco-Customer-Churn.csv
│   ├── chat_transcript.pdf
│   └── images/
└── Part_2/
    └── ...
```

其中 `Part_2` 现在甚至可以先不创建，等之后做 Part 2 再加。

## README 我们现在已经可以正式写了

而且 README 不需要你重新整理实验数字。我已经有你 notebook 里的真实结果：

*Logistic Regression — Accuracy 80.55%, Precision 65.72%, Recall 55.88%, F1 60.40%, ROC-AUC 84.21%*

以及 threshold、EDA、模型比较等真实结果，所以 README 可以完全基于你的实际实验，而不是模板。

README 最终会包含：

Assignment Overview → Dataset → AI-Assisted Workflow → Data Cleaning → EDA Findings → Models → Model Comparison → Selected Model → Threshold Analysis → Business Insights → Repository Structure → Medium Article → YouTube Demo → Chat Transcript → How to Run

这样老师打开 GitHub 首页基本就能看懂你做了什么。

下一步我们可以直接开始 GitHub Repository + 完整 README.md。我会给你一份可以直接放进 GitHub 的最终版本，同时告诉你 GitHub 上具体怎么创建目录和上传刚才这份 notebook。

□memcite□

#40

U 你

好的，我们开始github部分的任务吧

好，我们现在开始 **GitHub 部分**。先把 repository 和目录搭好，再写 README；Medium 和 YouTube 链接之后补进去即可。

目前这里没有可直接替你操作 GitHub 仓库的已连接工具，所以这一步你在 GitHub 网页操作，我逐步带你做。

## Step 1: 创建 GitHub Repository

打开 `url` GitHub `https://github.com/`，登录后右上角点：

+ → New repository

建议填写：

```
Repository name:
CMPE255-Assignment-1

Description:
CMPE 255 Assignment 1 - End-to-End AI-Assisted Data Science

Visibility:
Public
```

老师明确要求最终 GitHub 能 public view，所以这里直接选择：

Public

其他选项：

☒ Add a README file

`.gitignore` 和 License 暂时都可以不选。

然后：

Create repository

## Step 2: 建立 Assignment 结构

创建完成后，我们最终希望 repository 是：

```
CMPE255-Assignment-1/
├── README.md
├── Part_1/
│   ├── CMPE255_Assignment1_Telco_Churn.ipynb
│   ├── WA_Fn-UseC_-Telco-Customer-Churn.csv
│   └── chat_transcript.pdf
└── Part_2/
    └── ...
```

不过现在 `chat_transcript.pdf` 还没有，Part 2 也还没开始，所以现在只上传已有文件。

### Step 3: 上传 Part 1 Notebook 和 Dataset

GitHub 网页创建空文件夹不太方便，所以最简单的方法是：

在 repository 首页点击：

**Add file → Upload files**

然后把你的两个文件拖进去：

```
CMPE255_Assignment1_Telco_Churn.ipynb
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

但是这里有个小问题：我们希望它们进入 `Part_1/`。

GitHub 网页的 Upload files 对创建目标目录不总是最直观。因此更简单的学生做法是：

先点：

**Add file → Create new file**

文件名输入：

```
Part_1/README.md
```

GitHub 会自动创建：

```
Part_1/
```

然后里面先随便写：

```
# Part 1 - End-to-End AI-Assisted Data Science
```

点击：

**Commit changes**

现在 repository 就有：

```
README.md
```

```
Part_1/  
README.md
```

进入 `Part_1` 文件夹 → Add file → Upload files。

上传：

```
CMPE255_Assignment1_Telco_Churn.ipynb  
WA_Fn-UseC_-Telco-Customer-Churn.csv
```

Commit message 可以写：

```
Add Part 1 Telco churn analysis
```

然后：

Commit changes

---

## Step 4: 检查 Notebook 能不能在 GitHub 正常显示

上传以后，点击：

```
Part_1/CMPE255_Assignment1_Telco_Churn.ipynb
```

GitHub 应该自动 render Jupyter Notebook。

你应该可以直接往下看到：

### Project Overview

- Data Understanding
- Data Cleaning
- EDA 图表
- Logistic Regression
- Random Forest
- Gradient Boosting
- Model Comparison
- ROC Curve
- Feature Interpretation
- Threshold Analysis
- Business Recommendations
- Final Conclusion

如果所有图和 output 都能显示，说明最重要的 artifact 已经成功发布。

---

## Step 5: 主 README 怎么处理

现在 repository 根目录的:

```
README.md
```

才是老师进入 GitHub 后第一眼看到的东西。

它不能只是:

```
CMPE255 Assignment 1
```

而应该像一个小型 Data Science portfolio。

我建议最终结构:

```
# CMPE 255 Assignment 1
## End-to-End AI-Assisted Data Science

Overview

Assignment Deliverables

Part 1 – Customer Churn Prediction
  Dataset
  Business Problem
  AI-Assisted Workflow
  Data Cleaning
  Exploratory Data Analysis
  Machine Learning
  Model Comparison
  Selected Model
  Threshold Analysis
  Business Insights

Part 2 – Data Science Experiments
  [Tomorrow]

Repository Structure

How to Run

Medium Article
  [link]

YouTube Walkthrough
  [link]

ChatGPT Transcript
  [file]
```

最终 README 里还会放你真实的 model comparison:

| Model               | Accuracy | Precision | Recall | F1     | ROC-AUC |
|---------------------|----------|-----------|--------|--------|---------|
| Logistic Regression | 0.8055   | 0.6572    | 0.5588 | 0.6040 | 0.8421  |
| Random Forest       | 0.7835   | 0.6186    | 0.4813 | 0.5414 | 0.8206  |
| Gradient Boosting   | 0.8027   | 0.6655    | 0.5160 | 0.5813 | 0.8434  |

以及我们很值得展示的 threshold experiment:

*Lowering the Logistic Regression threshold from 0.50 to 0.30 increased churn recall from 55.88% to 75.40%.*

这会成为 README 一个很好的 highlight。

## 你现在先完成这一步

现在只做：

- ① 创建 Public repository: `CMPE255-Assignment-1`
- ② 创建 `Part_1/`
- ③ 上传 `.ipynb`
- ④ 上传 `.csv`

先不要自己编辑主 README，因为下一步我会直接给你一份完整可复制的 README，我们避免重复劳动。

完成后，把你的 GitHub repository 页面截图发给我。

然后我们下一步直接做：

*Step 6 — Final Professional README.md*

之后再处理 Medium → transcript → YouTube。

#42

U 你

已创建/上传，现在最外侧有一个part1文件夹和一个readme，part1文件夹中有上传的两个文件和一个readme

很好，这个结构现在是正确的：

```
CMPE255-Assignment-1/
├── README.md           ← 整个 Assignment 1 的主入口
└── Part_1/
    ├── README.md
    ├── CMPE255_Assignment1_Telco_Churn.ipynb
    └── WA_Fn-UseC_-Telco-Customer-Churn.csv
```

下一步就应该处理**最外层的 README.md**。不过因为 Part 2 还没做，Medium、YouTube 和 transcript 也还没有，我们现在不应该假装它们已经完成。

我建议主 README 现在先做成一个**可持续更新的最终框架**：Part 1 写完整，Part 2 标记待完成，链接之后补。

你可以把最外层 README.md 的内容整个替换为下面这版：

# # CMPE 255 Assignment 1 – AI-Assisted Data Science

## ## Overview

This repository contains my work for CMPE 255 Assignment 1.

The goal of this assignment is to use an AI coding assistant or chatbot to perform data science workflows end-to-end, understand the generated outputs, and document the complete process.

The repository is organized into two parts:

- **Part 1:** End-to-end AI-assisted data science using the Telco Customer Churn dataset.
- **Part 2:** Replication and exploration of data science experiments using prompts provided in the course repository.

---

## # Part 1 – Telco Customer Churn Prediction

### ## Project Goal

The goal of Part 1 is to build an end-to-end customer churn prediction workflow using the Telco Customer Churn dataset.

The project investigates the following question:

> Can customer characteristics, service usage, contract information, and billing behavior be used to identify customers who are at risk of churn?

The workflow was developed with AI assistance and includes data understanding, cleaning, exploratory data analysis, preprocessing, machine learning, model evaluation, model interpretation, and business recommendations.

---

### ## Dataset

The project uses the **Telco Customer Churn** dataset.

The dataset contains:

- **7,043 customer records**
- **21 original columns**
- Customer demographic information
- Service subscriptions
- Contract information
- Payment information
- Monthly and total charges
- Customer churn status

The target variable is:

``Churn``

where:

- `No` = customer remained with the company
- `Yes` = customer churned

The original target distribution was:

- **No Churn:** 5,174 customers (73.46%)
- **Churn:** 1,869 customers (26.54%)

This class imbalance motivated the use of evaluation metrics beyond accuracy.

---

## ## End-to-End Data Science Workflow

The project follows the workflow below:

1. Business and data understanding
2. Data quality assessment
3. Data cleaning
4. Exploratory data analysis
5. Feature preprocessing
6. Train/test split
7. Logistic Regression
8. Random Forest
9. Gradient Boosting
10. Multi-metric model evaluation
11. Confusion matrix analysis
12. ROC and ROC-AUC analysis
13. Feature interpretation
14. Classification threshold analysis
15. Business recommendations

---

## ## Data Cleaning

An important data quality issue was identified in the `TotalCharges` column.

Although `TotalCharges` represents a numerical value, it was originally stored as an object/string column.

Further investigation identified **11 blank `TotalCharges` values**. All 11 records belonged to customers with:

```
`tenure = 0`
```

These customers appear to be new customers who had not yet accumulated total charges.

The column was therefore converted to numeric format and the blank values were replaced with `0` rather than removing the customer records.

The cleaned dataset retained all **7,043 customers**.

---

## # Exploratory Data Analysis

### ## Churn Distribution

Approximately:

- **73.46%** of customers did not churn.
- **26.54%** of customers churned.

Because the target classes are imbalanced, accuracy alone was not used to determine model quality.

---

### ## Contract Type

Contract type showed a strong relationship with churn.

| Contract Type  | Churn Rate |
|----------------|------------|
| Month-to-month | 42.71%     |
| One year       | 11.27%     |
| Two year       | 2.83%      |

Month-to-month customers had substantially higher churn than customers with longer contracts.

---

### ## Customer Tenure

Average customer tenure was:

| Customer Group | Average Tenure |
|----------------|----------------|
| No Churn       | 37.57 months   |
| Churn          | 17.98 months   |

Customers who churned tended to have considerably shorter relationships with the company.

---

### ## Monthly Charges

Average monthly charges were:

| Customer Group | Average Monthly Charge |
|----------------|------------------------|
| No Churn       | \$61.27                |
| Churn          | \$74.44                |

Customers who churned tended to have higher monthly charges.

---

## ## Internet Service

Churn rates differed substantially across internet service types:

| Internet Service    | Churn Rate |
|---------------------|------------|
| DSL                 | 18.96%     |
| Fiber optic         | 41.89%     |
| No internet service | 7.40%      |

Fiber optic customers showed the highest churn rate.

---

## ## Payment Method

Electronic check customers showed the highest churn rate:

| Payment Method            | Churn Rate |
|---------------------------|------------|
| Electronic check          | 45.29%     |
| Mailed check              | 19.11%     |
| Bank transfer (automatic) | 16.71%     |
| Credit card (automatic)   | 15.24%     |

These relationships represent associations in the dataset and should not automatically be interpreted as causal relationships.

---

## # Machine Learning

Three classification models were trained and evaluated:

- Logistic Regression
- Random Forest
- Gradient Boosting

The same train/test split was used for all three models to provide a fair comparison.

Categorical variables were processed using one-hot encoding, while numerical variables were standardized where appropriate.

Preprocessing and model training were combined using scikit-learn pipelines to reduce the risk of data leakage.

---

## ## Model Comparison

| Model | Accuracy | Precision | Recall | F1 Score | ROC-AUC |
|-------|----------|-----------|--------|----------|---------|
| ---   | ---      | ---       | ---    | ---      | ---     |

```
| **Logistic Regression** | **0.8055** | 0.6572 | **0.5588** | **0.6040** | 0.8421 |
| Random Forest | 0.7835 | 0.6186 | 0.4813 | 0.5414 | 0.8206 |
| Gradient Boosting | 0.8027 | **0.6655** | 0.5160 | 0.5813 | **0.8434** |
```

No model achieved the best result on every metric.

Gradient Boosting achieved the highest ROC-AUC and precision, while Logistic Regression achieved the highest accuracy, recall, and F1-score.

For this project, **Logistic Regression** was selected as the primary model because identifying customers who actually churn is important for retention applications.

---

## ## Logistic Regression Results

The selected Logistic Regression model achieved:

- **Accuracy:** 80.55%
- **Precision:** 65.72%
- **Recall:** 55.88%
- **F1-score:** 60.40%
- **ROC-AUC:** 84.21%

The confusion matrix was:

```
| | Predicted No Churn | Predicted Churn |
|---|---:|---:|
| Actual No Churn | 926 | 109 |
| Actual Churn | 165 | 209 |
```

The model correctly identified 209 churned customers but missed 165 customers who actually churned.

These false negatives are particularly important in a customer retention application because the company would fail to identify those customers as being at risk.

---

## # Model Interpretation

The Logistic Regression model identified several characteristics associated with higher predicted churn, including:

- Fiber optic internet service
- Month-to-month contracts
- Higher total charges
- Electronic check payments
- Lack of online security
- Lack of technical support

Characteristics associated with lower predicted churn included:

- Longer customer tenure
- Two-year contracts

- DSL internet service
- Customers without internet service

These model relationships were broadly consistent with the earlier exploratory data analysis.

The coefficients describe associations learned by the model and should not be interpreted as proof of causation.

---

#### # Classification Threshold Experiment

The default classification threshold of `0.50` does not necessarily provide the best operating point for every business application.

Different Logistic Regression thresholds were therefore evaluated:

| Threshold | Precision     | Recall        | F1 Score      |
|-----------|---------------|---------------|---------------|
| 0.30      | 0.5193        | <b>0.7540</b> | <b>0.6150</b> |
| 0.35      | 0.5432        | 0.7059        | 0.6140        |
| 0.40      | 0.5682        | 0.6684        | 0.6143        |
| 0.45      | 0.6021        | 0.6150        | 0.6085        |
| 0.50      | <b>0.6572</b> | 0.5588        | 0.6040        |

Reducing the threshold from `0.50` to `0.30` increased recall from **55.88%** to **75.40%**.

This means the model can identify substantially more customers who eventually churn, but at the cost of lower precision and therefore more false-positive retention alerts.

The appropriate threshold should depend on the business cost of contacting customers compared with the cost of losing customers.

---

#### # Business Recommendations

Based on the exploratory analysis and machine learning results, several potential actions can be considered:

- Focus on newer customers**  
Customers with shorter tenure showed substantially higher churn risk.
- Encourage appropriate customers toward longer contracts**  
Month-to-month customers showed much higher churn than customers with one-year or two-year contracts.
- Investigate the fiber optic customer experience**  
Fiber optic customers showed relatively high churn and were also associated with higher predicted churn in the model.
- Investigate electronic check customers**  
Electronic check users showed substantially higher churn than customers using other payment methods.

5. **\*\*Use predicted churn risk for retention targeting\*\***

A lower classification threshold could be used when the business prefers identifying more at-risk customers even at the cost of additional false-positive alerts.

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## # Key Takeaways

This project demonstrates that end-to-end data science involves more than simply training a machine learning model.

Important decisions included:

- Understanding data quality issues before modeling
- Interpreting exploratory patterns
- Preventing data leakage
- Comparing multiple models
- Evaluating models using more than accuracy
- Understanding false positives and false negatives
- Interpreting model features carefully
- Connecting classification thresholds with business objectives

A more complex model did not automatically produce better results. Logistic Regression remained highly competitive and provided the best recall and F1-score among the three evaluated models at the default threshold.

---

## # Repository Structure

```
```text
```

```
CMPE255-Assignment-1/
```

```
|
|— README.md
|
|— Part_1/
|   |— README.md
|   |— CMPE255_Assignment1_Telco_Churn.ipynb
|   |— WA_Fn-UseC_-Telco-Customer-Churn.csv
|
|— Part_2/
|   |— To be completed
```

## How to Run Part 1

1. Open `Part_1/CMPE255_Assignment1_Telco_Churn.ipynb` in Google Colab or Jupyter Notebook.
2. Make sure `WA_Fn-UseC_-Telco-Customer-Churn.csv` is available to the notebook.
3. Update the dataset path if necessary.
4. Run all notebook cells from top to bottom.
5. Review the generated EDA, machine learning results, and conclusions.

---

## AI-Assisted Development

An AI coding assistant was used throughout the project to assist with:

- Planning the data science workflow
- Writing and explaining Python code
- Debugging notebook issues
- Designing exploratory analyses
- Selecting evaluation metrics
- Interpreting machine learning outputs
- Comparing models
- Exploring classification threshold trade-offs
- Translating technical results into business insights

The generated outputs were reviewed and interpreted throughout the workflow rather than being treated as results without explanation.

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## Part 2 — Data Science Experiments

Part 2 will replicate and explore the data science experiments provided in the course repository using an AI coding assistant.

**Status:** To be completed.

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## Medium Article

A Medium article documenting the end-to-end Part 1 data science workflow will be added here.

**Link:** To be added

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## YouTube Walkthrough

A video walkthrough explaining the end-to-end workflow, outputs, interpretations, and lessons learned will be added here.

**Link:** To be added

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## Chat Transcript

The AI-assisted development transcript used during Part 1 will be included in the repository.

**Status:** To be added

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## Final Deliverables

- [x] Part 1 end-to-end data science notebook
- [x] Dataset
- [x] Exploratory data analysis

- [x] Machine learning model comparison
- [x] Model interpretation
- [x] Classification threshold experiment
- [ ] Medium article
- [ ] Chat transcript
- [ ] YouTube walkthrough
- [ ] Part 2 experiments

### 然后修改 `Part\_1/README.md`

这个不用写得和主 README 一样长，否则重复很多。

把它改成：

```
```markdown id="jvbpnn"
# Part 1 – End-to-End AI-Assisted Data Science

This directory contains the artifacts for Part 1 of CMPE 255 Assignment 1.

## Project

**Telco Customer Churn Prediction**

The project performs an end-to-end data science workflow including data cleaning,
exploratory data analysis, preprocessing, machine learning, model comparison,
interpretation, threshold analysis, and business recommendations.

## Files

- `CMPE255_Assignment1_Telco_Churn.ipynb` – Complete executed Jupyter/Google Colab
notebook.
- `WA_Fn-UseC_-Telco-Customer-Churn.csv` – Dataset used for the analysis.

Additional artifacts, including the AI chat transcript, will be added to this directory.

For the complete project overview and results, see the main repository `README.md`.
```

然后 Commit message 可以写：

Add Part 1 project documentation

## 完成以后检查两个关键点

回到 repository 首页，你应该一打开就能看到完整的 README。然后点进：

Part\_1 → .ipynb

确认 GitHub 能正常 render notebook，特别检查：

EDA 图 → Model Comparison → Confusion Matrix → ROC Curve → Feature Coefficients → Final Conclusion

如果这些全部显示正常，那么 GitHub 主体已经搭好了。

接下来我建议我们马上处理 **ChatGPT transcript**，因为我们现在这整段对话本身就是 Part 1 的 agentic coding journey；把它保存下来后，再做 Medium article，最后根据 README + Medium 直接生成 YouTube 讲稿，会非常顺。□memcite□